Consequence-Aware Agentic AI: A Pre-Execution Outcome Assessment Framework for Responsible Large Language Model Agents

***Abstract*—Large language model (LLM) agents commit to generated action sequences without evaluating their downstream consequences before execution. We introduce *Consequence-Aware Agentic Artificial Intelligence* (CA^{2}I), which adds a *Consequence Assessment Module* (CAM) to the generate-then-execute paradigm: CAM simulates every candidate against a historical consequence model, admits only candidates that satisfy a hard risk constraint, and selects the survivor with the best predicted risk-adjusted outcome. We prove that, under stationarity, Sharpe-ratio selection dominates random selection in expectation, with a gap that grows with the candidate-set size K. We instantiate CA^{2}I for LLM-generated trading strategies on ten United States (US) equities. On a held-out window, CA^{2}I attains a mean Sharpe of 0.51 versus 0.37 for random selection, ahead of three portfolio-optimisation selectors, although differences at $n{=}10$ are not significant. A cross-validated ensemble variant, CA^{2}I$^+$, raises Sharpe to 0.77 versus 0.53 over 80 multi-regime cells (Wilcoxon $p{=}0.011$). Against an equal-weight (EW) ensemble of all candidates, consequence assessment provides two complementary advantages. First, because individual technical strategies are active on only 41.5% of trading days, static $1/K$ slicing leaves nearly 60% of capital idle in cash; an active-capital execution rule ($\mathrm{CA}^2\mathrm{I}^+_{\mathrm{act}}$) that allocates across currently open consequence-selected strategies raises rolling Sharpe to 0.92 versus 0.69 for EW (Wilcoxon $p{=}0.015$) and annualised return to 42.2% versus 21.7% ($p{=}2.5{\times}10^{-6}$), and lifts deployment Sharpe from 0.57 (EW) to 0.95. Second, CAM's cross-validated drawdown forecast (Spearman $r_S{=}0.37$, positive in $73/80$ cells) cuts rolling drawdown from -5.2% (EW) to -3.4% without cash abstention ($p{=}2.4{\times}10^{-12}$) and to -2.9% under a 7.5% risk budget ($p{=}5.8{\times}10^{-9}$), eliminating all cells above 10% drawdown. Unlike both deterministic text rankers and a neural LLM-as-a-judge (`Llama-3.1-8B-Instruct`), which an adversarial relabelling attack steers to the riskiest candidate in 100% of cells, CA^{2}I's selection is completely unaffected (0%). CA^{2}I thus offers an auditable pathway toward Responsible Artificial Intelligence (AI) for agentic systems.**

***Impact Statement*—Every large language model agent deployed today operates on the same unexamined assumption: that the first action sequence the model generates is safe to execute. This assumption is never validated—the agent acts, consequences follow, and if those consequences are harmful, the damage is already done. This paper breaks that assumption at the architectural level by introducing *Consequence-Aware Agentic AI* (CA^{2}I), a formally grounded framework that inserts a dedicated pre-execution consequence evaluation step—with a hard, simulation-based risk constraint—between an LLM's generation of candidate action sequences and their irreversible commitment to the environment. Where common agentic systems—ReAct, AutoGPT, Reflexion—either act immediately or reflect only after harm has occurred, CA^{2}I evaluates, compares, and constrains the predicted downstream consequences of all candidate sequences before a single action is taken. The practical significance is concrete: across ten diversified financial assets, this single architectural addition raises mean risk-adjusted return above that of random strategy execution and of established portfolio-optimisation selectors, reduces maximum drawdown relative to passive holding, and—unlike a return-maximising heuristic—does so without increasing trading turnover or sacrificing robustness to transaction costs. Wrapped around a diversified ensemble, the same module enforces a hard drawdown budget before execution, roughly halving realised drawdown at unchanged risk-adjusted return. We report these gains together with a candid statistical assessment (bootstrap confidence intervals and non-parametric tests) and a multi-regime rolling-window evaluation, so that the magnitude and the limits of the empirical evidence are both transparent. Beyond performance, CA^{2}I directly addresses the accountability gap that regulators and AI governance bodies have identified as the central unsolved problem in deployed agentic AI: the inability to audit why an agent acted as it did, whether the harm was foreseeable, and how it could have been prevented. CA^{2}I makes every selection decision mathematically traceable, audit-ready, and resistant to adversarial relabelling of proposals—which we verify experimentally—because its risk constraints operate on *simulated consequences*, not textual descriptions.**

***Index Terms*—Agentic AI, large language models, consequence assessment, responsible AI, sequential decision-making, risk-adjusted returns, Sharpe ratio, pre-execution evaluation, financial AI, AI safety.**

I. Introduction

THE rapid proliferation of large language model (LLM) agents in high-stakes sequential decision-making domains—from autonomous code execution [1], [2] to clinical pathway planning [3] and financial analysis [4]—has exposed a critical architectural gap: existing agentic frameworks generate action sequences and commit to execution without formally evaluating the *consequences* of those sequences in advance. This generate-then-execute paradigm is adequate when actions are reversible and low-stakes, but is fundamentally unsuitable when actions are irreversible, costly, or carry significant risk—precisely the conditions that define real-world deployment.

Consider the canonical ReAct [1] and AutoGPT [5], [6] architectures, which operate in a thought-action-observation loop in which the LLM proposes and immediately executes each action. Neither framework contains a formal mechanism for: (i) comparing alternative action sequences before execution; (ii) estimating the downstream outcome of the chosen sequence; or (iii) abandoning a sequence whose predicted consequences are unacceptable. The absence of such a mechanism creates a fragile pipeline in which the quality of the outcome depends entirely on whether the LLM's first-choice sequence happens to be sound—a brittle assumption, particularly in non-stationary environments or when the LLM's training distribution differs from the deployment domain.

Closing this gap requires more than a heuristic re-ranking of LLM outputs: it requires making pre-execution consequence

assessment a *first-class architectural component* whose decisions operate on simulated outcomes rather than on textual self-descriptions, so that they are auditable and resistant to prompt manipulation. This paper develops such a component and makes the following contributions:

1) **Framework.** We introduce *Consequence-Aware Agentic Artificial Intelligence (AI)* ($\mathrm{CA}^2\mathrm{I}$), a formal framework that decouples action-sequence *generation* from action-sequence *evaluation* by inserting an explicit Consequence Assessment Module (CAM) between generation and execution. CAM imposes a *hard, simulation-based risk constraint*—a property absent from vanilla LLM-agent pipelines—and we delineate precisely how $\mathrm{CA}^2\mathrm{I}$ differs from generate-and-rank, self-consistency, and classical look-ahead search (Sections II and III).
2) **Theory.** We formalise $\mathrm{CA}^2\mathrm{I}$ as a constrained sequential decision problem and prove that risk-adjusted selection dominates random selection in expectation under stationarity; we further quantify the dominance gap via order statistics and show it is non-decreasing in the candidate-set size K, and we bound the variance-reduction effect of selection (Section IV).
3) **Empirical evaluation with full robustness battery.** We instantiate and evaluate $\mathrm{CA}^2\mathrm{I}$ for LLM-driven trading-strategy selection on ten diversified United States (US) equities. Beyond naïve baselines, we benchmark against *established portfolio-optimisation selectors* (mean-variance, risk-parity, minimum-variance), a return-maximising heuristic, and a diversified ensemble; we report bootstrap confidence intervals and non-parametric tests, a transaction-cost sweep with a turnover-bias analysis, an eight-window multi-regime rolling evaluation, and a generation-sensitivity study. We disclose both the gains and their statistical limits candidly (Sections V–VI).
4) **An enhanced consequence model ($\mathrm{CA}^2\mathrm{I}^+$) with a statistically significant edge over random selection.** Diagnosing the base selector's reliance on a single, overfit-prone in-sample fit, we introduce a cross-validated, robustness-penalised, top-m ensemble consequence model (Section III-D). Pooled across eight market regimes it improves mean out-of-sample Sharpe over random selection by $+0.24$ (n=80; Wilcoxon p=0.011; bootstrap confidence interval (CI) excludes zero; positive in all 10 generation seeds) and over the base selector by $+0.21$ (Section VI-J).
5) **What consequence assessment adds beyond an ensemble and beyond generate-and-rank.** We dissect the comparison against an equal-weight (EW) ensemble of all K candidates and against text-based generate-and-rank pipelines, including a neural LLM-as-a-judge (`Llama-3.1-8B-Instruct`). First, because individual long/flat strategies are in the market on only 41.5% of trading days, static $1/K$ slicing leaves nearly 60% of capital idle in cash; pairing consequence-based top-m selection with active-capital execution ($\mathrm{CA}^2\mathrm{I}^+_{\mathrm{act}}$) raises rolling Sharpe to 0.92 versus 0.69 for EW (Wilcoxon p=0.015) and annualised return to 42.2% versus 21.7% ($p=2.5\times10^{-6}$), and lifts deployment Sharpe from 0.57 to 0.95. Second, CAM's cross-validated drawdown forecast (r_S=0.37, positive in $73/80$ cells) cuts rolling drawdown from -5.2% (EW) to -3.4% without cash abstention ($p=2.4\times10^{-12}$) and to -2.9% under a hard 7.5% risk budget ($p=5.8\times10^{-9}$). Third, under an adversarial relabelling attack, both deterministic text rankers and the neural LLM-as-a-judge select the camouflaged riskiest candidate in 100% of cells while incurring $\approx$ 1,000–1,300 extra tokens per cell, whereas CAM's selection changes in 0% of cells at zero token cost (Sections VI-K–VI-M).
6) **Responsible-AI operationalisation.** We show that $\mathrm{CA}^2\mathrm{I}$ is not merely a performance technique but a concrete, computationally tractable mechanism for *auditable* agentic decision-making: because every candidate's full consequence vector is logged and the selection rule is an explicit function of simulated outcomes, $\mathrm{CA}^2\mathrm{I}$ makes foreseeability, alternatives, and rationale traceable—directly operationalising requirements of the European Union Artificial Intelligence Act (EU AI Act) [7], the National Institute of Standards and Technology (NIST) AI Risk Management Framework (RMF) [8], and the Institute of Electrical and Electronics Engineers (IEEE) Ethically Aligned Design [9]. We argue this auditability, rather than raw return, is the framework's primary value (Section VII).

II. Related Work

A. LLM-Based Agentic Systems

The conceptual foundation for LLM agents traces to tool-use research [10], operationalised into end-to-end agents by ReAct [1]. Subsequent work scaled planning across multiple steps: AutoGPT [5] and BabyAGI decompose high-level goals into sequentially executed sub-tasks. Voyager [11] introduced a library of reusable agent skills, while Reflexion [12] added verbal self-reflection as a lightweight proxy for evaluating past actions. Multi-agent coordination frameworks such as CAMEL [13] and AutoGen [14] further extend these ideas to networks of specialised agents.

Despite this progress, these frameworks do not contain a formal mechanism for evaluating the consequences of a proposed action sequence *before commitment*. Reflexion approaches this most closely but operates post hoc: the agent reflects on what went wrong after the fact, rather than preventing harmful actions before they occur.

B. Planning, Search, and Look-Ahead

The value of evaluating consequences before acting has deep roots in classical AI planning [15]. Model-based reinforcement learning [16], [17] learns a world model for simulating outcomes, formalised in Dyna [18] and scaled to superhuman performance in MuZero [19]. More recently, Tree-of-Thoughts [20] brought structured multi-path search back to LLMs, and AlphaCode [21] employs large-scale candidate generation and filtering for code synthesis. $\mathrm{CA}^2\mathrm{I}$

shares this look-ahead philosophy but is distinguished by its generality, explicit risk-adjusted selection criterion, and orientation toward responsible deployment [22], [23].

C. Generate-and-Rank, Sampling, and Verification

A natural question is whether CA^{2}I is simply a *generate-and-rank* pipeline. It is instructive to make the distinction precise. Self-consistency [24] samples multiple chains of thought and aggregates by majority vote; best-of-n sampling with a learned verifier or reward model [25] re-scores sampled outputs; AlphaCode filters by execution against test cases. These methods rank or vote over *model-internal* signals (token likelihoods, learned reward scores, or pass/fail unit tests). CA^{2}I differs in three respects that matter for high-stakes deployment. *(i) Consequence model.* CA^{2}I ranks candidates by a quantitative, domain-grounded *simulation of downstream outcomes* on a held-out historical environment, not by textual self-description or a likelihood proxy; a strategy that *describes* itself as low-risk but exhibits large simulated drawdown is rejected regardless of its description. *(ii) Hard risk constraint.* Selection is subject to an explicit feasibility constraint $\rho(\cdot) \leq \rho_{\max}$ on a risk functional of the simulated outcome, so CA^{2}I can decline to act, rather than always returning the top-ranked sample. *(iii) Auditability.* The full consequence vector of every candidate is logged, making the decision foreseeable and traceable for governance. CA^{2}I is therefore best understood as a *constrained, consequence-grounded selection* layer that subsumes generate-and-rank as the special, unconstrained, proxy-scored case. Table I summarises the comparison along the dimensions that matter for deployment, and Section VI-L tests it empirically.

Test-time compute and its returns. Generate-and-rank is one instance of the broader strategy of spending more inference-time compute to improve LLM outputs. Repeated sampling raises the probability that *some* sample is correct roughly log-linearly in the number of samples [26], and compute-optimal allocation between sampling, revision, and verifier-guided search can outperform scaling model parameters [27]; process-level verifiers improve the selection step itself [28], and LLM-as-a-judge evaluation is a common, cheaper substitute for trained verifiers [29]. Recent evidence from agentic pipelines, however, shows sharply diminishing returns: in *The Token Tax*, Agrawal *et al.* [30] find that larger reasoning budgets in data-science agents raise token cost by more than 110% with negligible quality gains, and that correctness failures such as target leakage are removed not by more reasoning but by an explicit governance layer. CA^{2}I is complementary to this line of work in two ways. First, its selection signal is obtained from a domain simulator rather than from further LLM calls, so the cost of assessing a candidate is independent of the LLM's token budget (Section VI-L reports ≈ 40 ms of central processing unit (CPU) time and zero additional tokens per candidate). Second, it plays exactly the role of the external governance layer that [30] identify as necessary, but with a quantitative, auditable criterion. Because verifier and judge signals are computed from text, they are also exposed to prompt injection [31]; we test this exposure directly in Section VI-L.

D. LLMs in Financial Decision Support

BloombergGPT [4] demonstrated strong financial natural language processing (NLP) capability, and FinGPT [32] provided an open-source alternative. LLMs have been applied to sentiment-driven trading [33] and portfolio decision support [34]. None of these works employs pre-execution consequence assessment: they generate recommendations and act on them immediately or present them directly to a human without a formal evaluation step.

E. Responsible AI and AI Safety

The EU AI Act [7], the NIST AI RMF [8], and IEEE P7000 [9] all mandate transparency, accountability, and harm-awareness. Alignment research [35], [36] and constitutional AI [37] address value alignment at training time; red-teaming [38] probes deployed models post hoc. These mechanisms act at training or audit time; comparatively little work provides an *inference-time*, per-decision filter that is grounded in simulated consequences and auditable on a case-by-case basis—the niche the present framework targets, as developed in Sections III and VII.

III. The CA^{2}I Framework

A. Problem Formulation Grounded in the Trading Domain

We state the problem abstractly and then immediately ground every abstract object in the financial instantiation used throughout the paper.

State space $\mathcal{E}$. A state $s_t \in \mathcal{E}$ is the information available at decision epoch t. Concretely, s_t is the trailing window of daily open, high, low, close, and volume (OHLCV) records for a single asset up to and including day t, $s_t = \{(O_j, H_j, L_j, C_j, Vol_j)\}_{j \leq t}$, together with the technical indicators derived from it: moving averages (MAs), the relative strength index (RSI), Bollinger bands (BBs), and the moving average convergence/divergence (MACD) indicator. C_j denotes the split-adjusted closing price (Section V-A details the price basis).

Action and action sequence. An *action* $a_i \in \mathcal{A} = \{\text{ENTER}, \text{EXIT}, \text{HOLD}\}$ is a discrete position command executed at the closing price. An *action sequence* (equivalently, a *strategy*) τ is not an explicit list of per-day commands but a parameterised decision rule $\tau = (\theta, g_\theta)$, where $g_\theta : \mathcal{E} \to \mathcal{A}$ maps each state to an action and θ are the strategy parameters (e.g. the MA windows, the RSI thresholds, or the stop-loss level). Applying g_θ day by day over an evaluation window $\{t_0, \ldots, t_0 + T\}$ generates the realised command sequence $(a_{t_0}, \ldots, a_{t_0+T})$ and hence a position path $\pi_t \in \{0, 1\}$ (fully invested or in cash), which is what we mean by "the sequence induced by τ".

Portfolio path induced by τ. Let V_0 be the initial capital and $r_t = C_t/C_{t-1} - 1$ the asset's simple daily return. Executing τ produces the *portfolio value path* $\{V_t(\tau)\}_{t=t_0}^{t_0+T}$ defined recursively by

$$V_t(\tau) = V_{t-1}(\tau)\,\big(1 + \pi_{t-1}\, r_t\big), \qquad V_{t_0}(\tau) = V_0, \quad (1)$$

where $\pi_{t-1} \in \{0, 1\}$ is the position held entering day t as dictated by g_θ (with stop-loss overrides). This makes precise

TABLE I
SYSTEMATIC COMPARISON OF SELECTION PARADIGMS FOR LLM-GENERATED CANDIDATES. "TEXT-EXPOSED" MEANS THAT THE SELECTION SIGNAL IS A FUNCTION OF MODEL-GENERATED TEXT AND CAN THEREFORE BE ALTERED BY RELABELLING A CANDIDATE; "QUANTITATIVE RISK BUDGET" MEANS THAT A USER-SPECIFIED BOUND ON A RISK MEASURE OF THE OUTCOME IS ENFORCED BEFORE EXECUTION.

Paradigm	**Selection signal**	**Quantitative risk budget**	**Can abstain**	**Text-exposed**	**Per-proposal audit record**	**Extra cost per candidate**
Vanilla agent (ReAct) [1]	none (first sample executed)	no	no	—	no	none
Self-consistency [24]	majority vote over samples	no	no	yes	no	none
Best-of-n with verifier [25], [28]	learned reward of the output	no	no	yes	score only	reward-model call
LLM-as-a-judge [29]	judge's score of the text	no	no	yes	score only	judge LLM call
Execution filter (AlphaCode) [21]	unit-test pass/fail	binary only	partly	no	pass/fail	test execution
Equal-weight ensemble	none (every proposal executed)	no	no	no	no	none
CA^{2}I / CA^{2}I$^+$ (ours)	**simulated consequence vector**	**yes** ($\rho_{\max}$)	**yes**	**no**	**full vector**	**≈ 40 ms CPU, no tokens**

the phrase "portfolio path induced by τ" used in Definition 1. The strategy daily return is $y_t(\tau) = V_t(\tau)/V_{t-1}(\tau) - 1 = \pi_{t-1} r_t$.

Objective. Among a candidate set $\mathcal{T}$ generated for state s_t, the agent selects τ^* maximising a risk-adjusted utility $\mathcal{U}$ of the outcome subject to a risk constraint:

$$\tau^* = \arg\max_{\tau \in \mathcal{T}} \mathcal{U}\big(o(\tau, s_t)\big) \quad \text{s.t.} \quad \rho\big(o(\tau, s_t)\big) \le \rho_{\max}, \quad (2)$$

where the outcome $o(\tau, s_t) = \{V_t(\tau)\}$ is the simulated portfolio path, $\mathcal{U}$ is the Sharpe ratio, and ρ is the maximum drawdown (both defined next). Throughout the paper the symbol ρ is reserved for this risk functional and its budget $\rho_{\max}$.

B. Evaluation Criteria and Notation

Because these criteria are used by the selector and the theory, we define them before first use. *Notation convention.* Symbols without a hat denote *population* (expected) quantities of the return-generating process, and symbols with a hat denote their *sample* estimates computed from a finite simulated path. Let $\{y_t(\tau)\}_{t=1}^{T}$ be the strategy's daily returns over a window of T trading days. The *expected* daily return and volatility are $\mu(\tau) = \mathbb{E}[y_t(\tau)]$ and $\sigma(\tau) = \mathrm{sd}[y_t(\tau)]$; their sample counterparts are the sample mean and sample standard deviation

$$\begin{aligned} \hat{\mu}(\tau) &= \frac{1}{T}\sum_{t=1}^{T} y_t(\tau), \\ \hat{\sigma}(\tau) &= \Big(\frac{1}{T-1}\sum_{t=1}^{T}\big(y_t(\tau) - \hat{\mu}(\tau)\big)^2\Big)^{1/2}. \end{aligned} \quad (3)$$

Unless stated otherwise we use *simple* (arithmetic) returns $y_t = V_t/V_{t-1} - 1$; the logarithmic variant $y_t^{\log} = \ln(V_t/V_{t-1})$ yields qualitatively identical selection because the two agree to first order and our selection rule is monotone in the return level. The *annualised volatility* is $\hat{\sigma}_{\mathrm{ann}}(\tau) = \hat{\sigma}(\tau)\sqrt{T_{\mathrm{ann}}}$ with $T_{\mathrm{ann}} = 252$.

The *annualised Sharpe ratio* [39] is estimated by

$$\widehat{\mathrm{SR}}(\tau) = \frac{\hat{\mu}(\tau) - r_f}{\hat{\sigma}(\tau)} \cdot \sqrt{T_{\mathrm{ann}}}, \quad (4)$$

whose population counterpart is $\mathrm{SR}^*(\tau) = \sqrt{T_{\mathrm{ann}}}\,(\mu(\tau) - r_f)/\sigma(\tau)$, where r_f is the daily risk-free rate (Section V specifies its value and justification). The *maximum drawdown* of the portfolio path (1) is the largest peak-to-trough relative decline,

$$\mathrm{DD}(\tau) = \max_{t_0 \le s \le t \le t_0 + T} \frac{V_s(\tau) - V_t(\tau)}{V_s(\tau)}. \quad (5)$$

DD is a positive fraction (a 5% peak-to-trough decline is $\mathrm{DD} = 0.05$) and is reported as such in Table III; the multi-regime tables (Tables VII–IX) show the equivalent signed peak-to-trough return (-5%), so that "higher is better" holds for every column. When (5) is evaluated on a *simulated* path it is an estimate and is written $\widehat{\mathrm{DD}}(\tau)$. The *Calmar ratio* [40] relates annualised return to drawdown,

$$\widehat{\mathrm{Cal}}(\tau) = \frac{\hat{\mu}(\tau) \cdot T_{\mathrm{ann}}}{\widehat{\mathrm{DD}}(\tau)}, \quad (6)$$

and serves as a secondary, drawdown-aware selection criterion in the sensitivity analysis. Two further downside measures are reported in Section VI-K: the *Sortino ratio* [41], which replaces $\hat{\sigma}$ in (4) by the standard deviation of negative daily returns only, and the 5% *conditional value-at-risk* ($\mathrm{CVaR}_{5\%}$) [42], the mean of the worst 5% of daily returns.

Definition 1 (Consequence vector)**.** *The* consequence *of strategy τ from state s_t is a d-dimensional vector*

$$\begin{aligned} c(\tau, s_t) = \big(&\hat{\mu}(\tau),\, \hat{\sigma}(\tau),\, \widehat{\mathrm{SR}}(\tau),\, \widehat{\mathrm{DD}}(\tau),\, \widehat{\mathrm{Cal}}(\tau)\big) \\ &\in \mathbb{R}^d, \quad d = 5, \end{aligned}$$

where $\hat{\mu}(\tau)$ is the sample mean daily return (3) *(an estimate of the expected return $\mu(\tau)$), and the remaining components are given by* (3)–(6)*, all evaluated on the path simulated by the consequence model. We deliberately define the consequence as a real vector (not a tuple of heterogeneous objects) so that it coincides with the codomain $\mathbb{R}^d$ of the Consequence Assessment Module in Definition 2; the expected outcome, its dispersion, and the drawdown are simply designated components of this vector.*

Definition 2 (Consequence Assessment Module)**.** *The* Consequence Assessment Module *(CAM) is the function $\phi : \mathcal{T} \times \mathcal{E} \to \mathbb{R}^d$ that maps a candidate strategy and the current state to its consequence vector $c(\tau, s_t)$ of Definition 1. CAM is*

implemented by simulating each candidate against a historical consequence model $\mathcal{M}_{\text{hist}}$ (an event-driven backtester over $\mathcal{W}_{\text{train}}$) and computing the empirical components of $c(\tau, s_t)$.

C. Framework Architecture

CA^{2}I comprises four modules operating in sequence at each decision epoch:

(M1) LLM Strategy Generator ($\mathcal{G}$). Given state s_t and a structured prompt $p(s_t)$, the LLM generates a set $\mathcal{T} = \{\tau_1, \tau_2, \ldots, \tau_K\}$ of K candidate action sequences. Each τ_k is a fully specified, executable plan with entry conditions, exit conditions, and risk parameters.

(M2) Consequence Assessment Module (CAM, ϕ). For each $\tau_k \in \mathcal{T}$, CAM simulates the sequence against the historical model $\mathcal{M}_{\text{hist}}$ and computes the consequence vector:

$$\phi(\tau_k) = \Big(\hat{\mu}_k,\ \hat{\sigma}_k,\ \widehat{\text{SR}}_k,\ \widehat{\text{DD}}_k,\ \widehat{\text{Cal}}_k\Big), \tag{7}$$

where the subscript k abbreviates the argument τ_k: $\hat{\mu}_k = \hat{\mu}(\tau_k)$ is the sample mean daily return, $\hat{\sigma}_k$ the sample return volatility, $\widehat{\text{SR}}_k$ the annualised Sharpe ratio, $\widehat{\text{DD}}_k$ the maximum drawdown, and $\widehat{\text{Cal}}_k$ the Calmar ratio, all estimated on $\mathcal{W}_{\text{train}}$.

(M3) Consequence-Aware Selector ($\mathcal{S}$). The selector applies the risk-adjusted criterion:

$$\tau^* = \mathcal{S}\big(\{\phi(\tau_k)\}_{k=1}^K\big) = \tau_{k^*}, \quad k^* = \arg\max_k\ \widehat{\text{SR}}_k, \tag{8}$$

subject to $\widehat{\text{DD}}_{k^*} \le \rho_{\max}$ when an explicit drawdown constraint is imposed.

(M4) Executor ($\mathcal{X}$). Only the selected sequence τ^* is committed to the environment. All other candidates are discarded without execution.

The full pipeline is:

$$s_t \xrightarrow{\mathcal{G}} \mathcal{T} \xrightarrow{\phi} \{\phi(\tau_k)\} \xrightarrow{\mathcal{S}} \tau^* \xrightarrow{\mathcal{X}} s_{t+1}. \tag{9}$$

D. Robust Consequence Assessment: Cross-Validated Ensemble Selection (CA^{2}I$^+$)

The base selector (8) estimates each consequence statistic from a *single* contiguous in-sample fit. As our experiments confirm (Section VI), a single window is a high-variance, upward-biased estimator of out-of-sample skill—the candidate that looks best in-sample is frequently the one that overfit noise (the selection "winner's curse" [43]). We therefore introduce an enhanced consequence model, **CA^{2}I$^+$**, that keeps the architecture (9) but replaces the point estimate with a *cross-validated, robustness-penalised, ensemble* assessment. Three principled changes, with all hyper-parameters fixed *a priori* (never tuned on test data), define it:

(i) Walk-forward cross-validation. The training window is partitioned into M contiguous out-of-sample blocks with an expanding warm-up [44]; each candidate is re-simulated on every block, yielding a *distribution* of fold Sharpe ratios $\{\widehat{\text{SR}}_k^{(j)}\}_{j=1}^M$ and fold drawdowns $\{\widehat{\text{DD}}_k^{(j)}\}$, rather than one in-sample value.

(ii) Robustness-penalised score (lower confidence bound). Candidates are ranked by

$$s_k \;=\; \overline{\text{SR}}_k \;-\; \lambda\,\text{std}_j\big(\widehat{\text{SR}}_k^{(j)}\big), \qquad \overline{\text{SR}}_k = \frac{1}{M}\sum_{j=1}^{M} \widehat{\text{SR}}_k^{(j)}, \tag{10}$$

a lower-confidence-bound (LCB) score that penalises strategies whose edge is unstable across sub-periods, subject to the hard feasibility filter $\text{median}_j\ \widehat{\text{DD}}_k^{(j)} \le \rho_{\max}$ (the cross-validated counterpart of the constraint in (2)).

(iii) Top-m ensemble and active-capital execution. Instead of committing to a single pick, CA^{2}I$^+$ selects the m highest-scoring feasible candidates $\mathcal{T}_m^* \subset \mathcal{T}$, diversifying away the idiosyncratic selection error that dominates at short horizons. We evaluate two causal execution rules in the Executor ($\mathcal{X}$, Module M4):

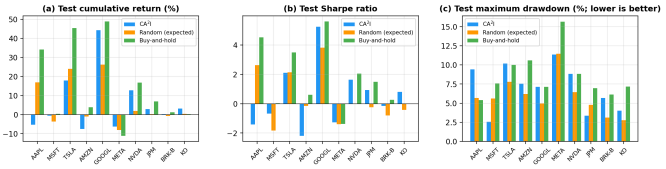
1) *Static $1/m$ allocation* (CA^{2}I$^+$): each selected candidate receives a fixed $1/m$ capital slice, giving daily portfolio return $y_t = \frac{1}{m}\sum_{k\in\mathcal{T}_m^*} y_t(\tau_k)$.
2) *Active-capital allocation* (CA^{2}I$^+_{\text{act}}$): because individual long/flat technical strategies are only invested on a fraction of trading days (41.5% on average in our rolling study) and sit in cash ($y_t(\tau_k)$=0) otherwise, static $1/K$ or $1/m$ slicing leaves the majority of capital idle on an average day. Under active-capital execution, capital on day t is divided equally among whichever selected candidates currently hold an open position, $\mathcal{A}_t = \{k \in \mathcal{T}_m^* : w_t(\tau_k) > 0\}$, yielding $y_t^{\text{act}} = \frac{1}{|\mathcal{A}_t|}\sum_{k\in\mathcal{A}_t} y_t(\tau_k)$ when $\mathcal{A}_t \neq \emptyset$ and 0 (cash) when no selected strategy is active. Because $w_t(\tau_k)$ is determined by signals up to day $t-1$, this rule is strictly causal and introduces zero look-ahead bias.

We use M=5, λ=0.5, $\rho_{\max}$=0.30, and m=3 throughout, except in the risk-budget study of Section VI-K, where $\rho_{\max}$ is varied explicitly and where we also evaluate ranking by cross-validated drawdown ($s_k^{\text{DD}} = -\text{median}_j\ \widehat{\text{DD}}_k^{(j)}$, denoted CA^{2}I$^+_{\text{MinDD}}$) or expected return (CA^{2}I-Return). Note that the equal-weight ($1/K$) ensemble of all candidates is the special case m=K, $\rho_{\max}$=∞ of static CA^{2}I$^+$, i.e. with consequence assessment switched off. CA^{2}I$^+$ preserves every auditability and hard-constraint property of CA^{2}I—the fold scores (10) and the ensemble membership are fully logged—while, as Sections VI-J and VI-K show, delivering statistically significant, regime-robust improvements over random selection, the base selector, and the equal-weight ensemble.

E. Instantiation in Financial Trading

In the financial domain, state s_t encodes the trailing price history, volume, and technical indicators. The LLM generates K=5 candidate strategies drawn from five canonical, long-only technical-analysis families that are widely used as multi-indicator trading rules [45]. Their decision rules g_θ are fixed and auditable, with the LLM choosing only the parameters θ (and a descriptive name; see Table II):

- **Moving-average crossover (MA):** enter when the short-window simple moving average rises above the long-



(d) Aggregate summary (ten equities)

Metric	CA²I	Random (expected)	Buy-and-hold
Mean cumulative return (%)	+6.12	+5.59	+14.64
Mean Sharpe ratio	+0.51	+0.37	+1.67
Mean maximum drawdown (%)	7.01	5.88	8.54

(e) Strategy selected by the Consequence Assessment Module, with its strategy family

Ticker	Selected strategy (GPT-5.4 name)	Family	Train Sharpe	Test return	B&H return
AAPL	Volatility Expansion Breakout	BB	0.63	-5.29%	+34.10%
MSFT	Overheat Pullback Reclaimer	RSI	0.77	-0.62%	+0.31%
TSLA	Acceleration Pulse MACD	MACD	0.88	+17.88%	+45.43%
AMZN	E-Commerce Thrust Runner	MOM	0.15	-7.55%	+3.78%
GOOGL	Golden Glide Trend Capture	MA	0.55	+44.32%	+48.83%
META	Trend Engine 21/100	MA	1.24	-6.37%	-11.14%
NVDA	AI Trend Rider Cross	MA	1.29	+12.72%	+16.79%
JPM	Banker Pullback Rebound	RSI	1.00	+2.82%	+6.94%
BRK-B	Capital Compound MACD Turn	MACD	0.76	+0.16%	+1.20%
KO	Staples MACD Reacceleration	MACD	0.25	+3.14%	+0.17%

Families: MA = moving-average crossover; RSI = relative strength index mean reversion; BB = Bollinger-band breakout; MACD = moving-average convergence/divergence signal; MOM = price momentum.

Fig. 1. CA^{2}I summary dashboard for the held-out deployment window (August–October 2025). (a)–(c) Per-stock test cumulative return, Sharpe ratio, and maximum drawdown for CA^{2}I, the expected value of uniform random selection, and buy-and-hold (B&H). (d) Means over the ten equities (identical to Table III). (e) Strategy-selection table: the name generated by the LLM, OpenAI's Generative Pre-trained Transformer GPT-5.4 (Section V), for the strategy selected by CAM, the *strategy family* to which it belongs (MA: moving-average crossover; RSI: relative strength index mean reversion; BB: Bollinger-band breakout; MACD: moving average convergence/divergence signal; MOM: price momentum), its training-period Sharpe ratio, and its test return next to the B&H return. The family of every one of the 50 generated candidates is given in Table II.

window average (a "golden cross"), exit on the reverse crossover or stop-loss. Captures sustained trends.

- **RSI mean-reversion (RSI):** enter when the Relative Strength Index falls below an oversold threshold, exit when it rises above an overbought threshold, after a holding cap, or on stop-loss. Exploits short-term overreactions.
- **Bollinger-band breakout (BB):** enter when price closes above the upper band (the rolling mean of the closing price plus k rolling standard deviations), exit on reversion to the middle band (the rolling mean) or stop-loss. Captures volatility-expansion breakouts.
- **MACD signal (MACD):** enter on a bullish crossover of the MACD line (the difference between a fast and a slow exponential moving average) above its signal line (an exponential moving average of the MACD line), exit on the bearish crossover or stop-loss. A momentum-confirmation rule.
- **Price momentum (MOM):** enter when the trailing L-day return exceeds a threshold, ride with a trailing stop and a maximum holding period. Participates in intermediate-term strength, in the spirit of cross-sectional momentum effects [46].

Each candidate is fully specified by its type and parameters (indicator windows, thresholds, and a stop-loss level). CAM backtests every candidate over $\mathcal{W}_{\text{train}}$ to estimate $\phi(\tau_k)$, and the executor applies only the selected strategy to the strictly held-out $\mathcal{W}_{\text{test}}$.

From descriptive names to families. The LLM is asked to give each candidate a short descriptive name (for example, "Golden Glide Trend Capture"), but the name is purely cosmetic: the executable rule is determined by the `type` field of the generated JavaScript Object Notation (JSON) object, which must be one of the five families above, and by its numeric parameters. Table II gives the complete mapping from every generated name to its family, with the candidate selected by CAM marked in bold; the selection table of Fig. 1 repeats the family of each selected strategy.

IV. Theoretical Analysis

A. The Risk-Adjusted Selection Criterion

CA^{2}I selects on the annualised Sharpe ratio (4) rather than on raw return because it explicitly penalises volatility: a high-return, high-variance strategy is downranked relative to a moderate-return, low-variance one—the correct preference under risk aversion. We treat the training-period Sharpe $\widehat{\mathrm{SR}}_k$ as the consequence statistic on which selection acts and analyse its out-of-sample behaviour below.

B. Dominance of Consequence-Aware Selection

Theorem 1 (CA^{2}I Dominance Under Stationarity)**.** *Let $\mathcal{T} = \{\tau_1, \ldots, \tau_K\}$ be a set of K candidate action sequences whose training-period Sharpe ratios $\{\widehat{\mathrm{SR}}_k\}$ are unbiased estimators of the corresponding test-period Sharpe ratios $\{\mathrm{SR}^*_k\}$ under stationarity. Define the CA^{2}I selector $k^* = \arg\max_k \widehat{\mathrm{SR}}_k$ and the random selector $k^R \sim \mathrm{Uniform}\{1, \ldots, K\}$. Then:*

$$\mathbb{E}\big[\mathrm{SR}^*_{k^*}\big] \;\geq\; \mathbb{E}\big[\mathrm{SR}^*_{k^R}\big] = \frac{1}{K}\sum_{k=1}^{K} \mathrm{SR}^*_k. \tag{11}$$

Proof: Under stationarity, $\mathbb{E}[\widehat{\mathrm{SR}}_k] = \mathrm{SR}^*_k$ for all k. The CA^{2}I selector chooses $k^* = \arg\max_k \widehat{\mathrm{SR}}_k$. For any collection of random variables $\{X_k\}_{k=1}^K$ and a selector $k^* = \arg\max_k X_k$:

$$\mathbb{E}[X_{k^*}] = \mathbb{E}\left[\max_k X_k\right] \;\geq\; \frac{1}{K}\sum_{k=1}^{K} \mathbb{E}[X_k],$$

where the inequality follows from the maximum being at least as large as the mean. Substituting $X_k = \widehat{\mathrm{SR}}_k$ and applying $\mathbb{E}[\widehat{\mathrm{SR}}_k] = \mathrm{SR}^*_k$ gives (11). ∎

Remark 1 (Honest scope of the guarantee)**.** *The stationarity assumption is idealised; real financial returns exhibit regime changes, volatility clustering, and structural breaks [47], [48], so $\mathbb{E}[\widehat{\mathrm{SR}}_k] = \mathrm{SR}^*_k$ holds only approximately. Theorem 1 therefore establishes a* principled expected-case baseline*, not a finite-sample guarantee. Two consequences follow. First, the realised benefit is governed by the bias $b_k = \mathbb{E}[\widehat{\mathrm{SR}}_k] - \mathrm{SR}^*_k$ induced by non-stationarity; when a regime shift makes the candidate with the largest in-sample Sharpe the worst out-of-sample (as we observe for the 2025Q1 window, i.e. the first quarter of 2025, in Section VI), the selection can underperform random. Second, selecting the maximum of K noisy estimates incurs selection bias [43], so the in-sample Sharpe of the winner overstates its true Sharpe. We therefore report out-of-sample results only, and we assess significance with multiple-comparison-aware, non-parametric, and bootstrap procedures in Section VI.*

TABLE II

MAPPING OF THE 50 GPT-5.4-GENERATED STRATEGY NAMES TO THEIR STRATEGY FAMILIES (ONE CANDIDATE PER FAMILY AND ASSET; TICKER SYMBOLS AND COMPANIES ARE LISTED IN SECTION V-A). BOLD: CANDIDATE SELECTED BY CAM (BASE CA^{2}I, MAXIMUM TRAINING SHARPE); THE NUMBER IN PARENTHESES IS ITS TRAINING-PERIOD SHARPE RATIO. NAMES ARE REPRODUCED VERBATIM FROM THE LLM OUTPUT (EV: ELECTRIC VEHICLE; AWS: AMAZON WEB SERVICES).

Ticker	MA crossover	RSI mean reversion	Bollinger-band breakout	MACD signal	Price momentum
AAPL	Golden Recovery Cross	RSI Snapback Accumulator	**Volatility Expansion Breakout (0.63)**	Signal Line Turnaround	Measured Momentum Ride
MSFT	Trend Rider 20/100	**Overheat Pullback Reclaimer (0.77)**	Upper Band Expansion Breakout	MACD Pulse Continuation	30-Day Surge with Trailing Defense
TSLA	EV Trend Reclaim Crossover	Volatility Snapback RSI Fade	High-Beta Band Expansion Breakout	**Acceleration Pulse MACD (0.88)**	Adaptive EV Momentum Run
AMZN	Prime Trend Glide	AWS Pullback Reclaimer	Marketplace Volatility Expansion	Cloud Signal Accelerator	**E-Commerce Thrust Runner (0.15)**
GOOGL	**Golden Glide Trend Capture (0.55)**	Alphabet RSI Snapback	Band Expansion Breakout Rider	Cloud Pulse MACD Trend Sync	Relentless Relative Strength Run
META	**Trend Engine 21/100 (1.24)**	RSI Pullback Snapback	Upper Band Expansion Breakout	MACD Acceleration Rider	30-Day Surge with Trailing Shield
NVDA	**AI Trend Rider Cross (1.29)**	Overheat Fade Snapback	Volatility Expansion Launch	Semiconductor Pulse MACD	Runaway Leader Momentum Stop
JPM	Golden Trend Rider	**Banker Pullback Rebound (1.00)**	Upper Band Expansion Surge	Institutional MACD Acceleration	Quarterly Strength With Trailing Defense
BRK-B	Golden Value Drift	Buffett Bounce Reversion	Range Expansion Breakout	**Capital Compound MACD Turn (0.76)**	Steady Giant Momentum Ride
KO	Defensive Trend Resume Crossover	Consumer Staple RSI Snapback	Quiet Range Volatility Expansion	**Staples MACD Reacceleration (0.25)**	Measured Breakout With Trailing Protection

C. The Dominance Gap and Its Dependence on K

Theorem 1 states only that selection is *at least* as good as the mean. The following proposition quantifies the gap and explains the empirical finding (Section VI) that CA^{2}I improves as the candidate-set size grows.

Proposition 1 (Selection gap and variance reduction)**.** *Let $X_1, \dots, X_K$ be the test-period Sharpe ratios of the candidates with $X_k = \mathrm{SR}_k^* + \varepsilon_k$, where the estimation noise ε_k has zero conditional mean. Define the selection gap $\Delta_K = \mathbb{E}[\max_k X_k] - \frac{1}{K}\sum_k \mathbb{E}[X_k] \geq 0$. Then* (i) *$\Delta_K$ is non-decreasing in K; and* (ii) *if the X_k are independent and identically distributed (i.i.d.) with standard deviation σ_X, then $\Delta_K \geq c_K \sigma_X$ with c_K strictly increasing in K (e.g. $c_K = \mathbb{E}[\max_k Z_k]$ for standard-normal Z_k, giving $c_2 \approx 0.56$, $c_5 \approx 1.16$, $c_{10} \approx 1.54$). Hence a larger or more dispersed candidate set yields a larger expected advantage of consequence-aware selection over random selection.*

Proof: (i) Adding a candidate can only raise the maximum: $\max_{k\leq K+1} X_k \geq \max_{k\leq K} X_k$ pointwise, so $\mathbb{E}[\max_{k\leq K+1} X_k] \geq \mathbb{E}[\max_{k\leq K} X_k]$, while the average of means is unchanged in expectation under exchangeability; thus $\Delta_{K+1} \geq \Delta_K$. *(ii)* For i.i.d. $X_k = \mu_X + \sigma_X Z_k$, $\mathbb{E}[\max_k X_k] = \mu_X + \sigma_X \mathbb{E}[\max_k Z_k]$ and the mean is μ_X, so $\Delta_K = \sigma_X \mathbb{E}[\max_k Z_k]$; the expected maximum of K i.i.d. mean-zero unit-variance variables is strictly increasing in K. ∎

Remark 2 (Variance of the selected outcome)**.** *Beyond raising the mean, selection concentrates outcomes: empirically (Section VI) the dispersion of CA^{2}I's selected Sharpe is no larger than that of random selection, because systematic suppression of the worst candidates trims the lower tail. This is the mechanism behind CA^{2}I's worst-case (downside) protection.*

D. Maximum Drawdown as a Hard Risk Constraint

By restricting execution to strategies satisfying $\widehat{\mathrm{DD}}_k \leq \rho_{\max}$ with DD defined in (5), CAM guarantees that no candidate whose historical drawdown exceeds the threshold is ever executed. This constitutes a *hard risk constraint at the architectural level*—a property absent from vanilla LLM-agent and generate-and-rank frameworks, which return their top-ranked sample unconditionally.

V. EXPERIMENTAL SETUP

A. Data: Asset Selection, Source, and Preprocessing

Asset-selection criteria. The ten equities were fixed *before* any strategy was generated or backtested, using four criteria that do not depend on any outcome measured in this study: (i) *liquidity and data integrity*—US-listed common stocks that are constituents of the Standard & Poor's 500 (S&P 500) index with a mean daily traded value of at least \$0.9 billion over the training window, so that close-to-close execution and negligible market impact are realistic assumptions; (ii) *continuous history*—listed and trading without interruption since before January 2021, so that every asset has the same 1,213-day sample; (iii) *sector coverage*—the set spans five Global Industry Classification Standard (GICS) sectors: information technology (Apple, AAPL; Microsoft, MSFT; NVIDIA, NVDA), consumer discretionary (Tesla, TSLA; Amazon, AMZN), communication services (Alphabet, GOOGL; Meta Platforms, META), financials (JPMorgan Chase, JPM; Berkshire Hathaway Class B, BRK-B), and consumer staples (Coca-Cola, KO); and (iv) *volatility coverage*—annualised training-period volatility ranges from 16.0% (KO) and 17.5% (BRK-B) through 24.7–35.3% (JPM, MSFT, AAPL, GOOGL, AMZN) to 44.5–62.0% (META, NVDA, TSLA), so that the consequence model is tested on both defensive and highly volatile price processes. The diversity is deliberate: a consequence assessment framework that

succeeds only on homogeneous high-growth equities would not constitute a general contribution. Because all ten are large, liquid survivors of the sample period, the set is subject to survivorship bias with respect to the broader market; this affects all methods equally and is discussed under Limitations.

Data source. Daily OHLCV records were downloaded from Yahoo Finance through the open-source `yfinance` Python library (`Ticker.history` with `auto_adjust=False`) for January 1, 2021 to October 31, 2025; the raw files, the download timestamp, and a per-ticker quality report are released with the code. *Price basis:* Yahoo Finance adjusts the open, high, low, and close fields for stock splits; we use this split-adjusted close for signals and returns, and retain the separate dividend-adjusted close only for verification. Strategy and buy-and-hold returns are therefore *price* returns (dividends are not reinvested), which is the same for every method and so cannot change any comparison. The effect is small: the daily-return correlation between the two price series is at least 0.995 for every ticker, and the annualised dividend drag is 0.0–0.8% for eight tickers and 2.7% and 3.0% for JPM and KO.

Preprocessing pipeline. Each raw series passes through the same deterministic pipeline: (1) *validation*—counts of missing values in every field, zero-volume days, duplicate dates, calendar gaps longer than five days, single-day moves exceeding $\pm 50\%$, and negative prices; (2) *cleaning*—removal of duplicate dates, chronological sorting, forward-filling of at most three consecutive missing values, and removal of any remaining incomplete rows (no interpolation or smoothing is applied, to avoid look-ahead bias); (3) *date-based splitting* into the windows below; and (4) *causal feature construction*—every indicator at day t uses only prices up to t, and indicators on the test window are initialised with a 250-day warm-up taken from the end of the training window, so that long look-back rules (e.g. a 200-day MA) are defined from the first test day without using any test-window data. All ten series passed validation with a quality score of 1.0: $1{,}213$ trading days each, no missing values, no zero-volume days, no duplicate dates, no gaps, and no anomalous moves, so the cleaning step removed no rows.

Train/test partition. The data are partitioned into two non-overlapping windows:

- **Training window** $\mathcal{W}_{\text{train}}$: January 4, 2021 – July 31, 2025 ($1{,}149$ trading days). Used exclusively for strategy generation context and consequence assessment (backtesting). No test-window data leak into this phase.
- **Test window** $\mathcal{W}_{\text{test}}$: August 1, 2025 – October 30, 2025 (64 trading days). Strictly held out; no strategy parameter was tuned on this window.

The multi-regime study (Section V-G) applies the same pipeline to eight further quarterly test windows, each with its own training history that ends where its test window begins.

Risk-free rate. We set the annual risk-free rate to $r_f^{\text{ann}} = 2\%$, i.e. a daily rate $r_f = 0.02/252$, as a fixed proxy for the short-term US Treasury bill (T-bill) yield [49]; over 2021–2025 the 3-month T-bill yield averaged in the low-single-digit percent range, and 2% is a conventional, mildly conservative central value. We use a fixed rate (rather than a contemporaneous daily series) so that Sharpe ratios are comparable across assets and windows, and because the test window is short. Critically, the risk-free rate enters every method identically and therefore cannot affect the *relative* ranking of CA^{2}I against the baselines; we verified that perturbing r_f^{ann} over $[0\%, 5\%]$ leaves all selection decisions and the sign of every pairwise comparison unchanged.

B. LLM Strategy Generator

The generator $\mathcal{G}$ is OpenAI GPT-5.4 (`gpt-5.4-2026-03-05`), the most capable reasoning model available at the time of experimentation, accessed via the chat completions application programming interface (API). For each equity, $\mathcal{G}$ receives a single structured prompt containing: (i) the ticker and company name; (ii) training-window market context (total and annualised return, annualised volatility, historical Sharpe, maximum drawdown, last price, 52-week high/low, 30- and 90-day returns, the 50- and 200-day simple moving averages SMA(50) and SMA(200), and the 14-day RSI); and (iii) a JSON schema requesting exactly $K{=}5$ strategies, one of each canonical type (MA, RSI, BB, MACD, MOM), each with typed numeric parameters, entry/exit rules, a stop-loss threshold, a name, and a free-text rationale. The system message constrains the model to emit only valid JSON, which is parsed and schema-validated; any candidate failing validation is replaced by a deterministic rule-based strategy of the same type. We use the API default decoding settings and fix the global seed to 42; the exact prompt template and all raw model outputs are released with the code. A fully deterministic rule-based generator is provided so that every result is reproducible offline *without* API access, and it is the generator used for the multi-regime and generation-sensitivity studies below.

We emphasise that the LLM's role is confined to *proposing* parameterised candidates; the executable decision logic g_θ and the consequence model are fixed and auditable. The theoretical contribution is not the strategies themselves but the constrained, consequence-grounded selection layer and its analysis (Section IV); the LLM is one interchangeable realisation of the generator $\mathcal{G}$.

C. Consequence Assessment via Backtesting

CAM implements an event-driven backtester on daily OHLCV data. Positions are entered and exited at the (split-adjusted) closing prices on the signal day. All strategies are simulated with initial capital $V_0 = \$100{,}000$. The frictionless setting (per-side cost $\kappa = 0$) is the default for the headline comparison so that no method is implicitly favoured; the effect of realistic commissions and slippage is studied explicitly in Section V-F rather than left as a caveat. The backtester records a complete trade log for each strategy (entry/exit dates and prices, return, profit and loss (P&L), holding days, exit reason), from which the consequence vector $\phi(\tau_k)$ of (7) is computed.

D. Baselines

Because random selection and buy-and-hold alone are weak benchmarks, we evaluate CA^{2}I against a spectrum of selectors,

all operating on *training-only* information and realised on the held-out test window, so that every method solves the identical task.

Naïve references. *Random (RND)*: uniform selection among the K candidates. We report its *expected* value (the mean over candidates, equal to the limit of infinitely many draws) and also draw 100 seed-controlled empirical samples per asset for the bootstrap. *Buy-and-Hold (B&H)*: purchase on the first test day and hold.

Established portfolio-optimisation selectors [50]–[52]. Treating the K candidate-strategy daily-return streams as assets, estimated on the training window: *Mean-Variance (MV)*, the long-only max-Sharpe (tangency) weights with a ridge-regularised covariance; *Risk-Parity (RP)*, inverse-volatility weights; *Minimum-Variance (Min-Vol)*, the lowest-volatility candidate; and the *Equal-weight (EW) ensemble* $(1/K)$, a no-selection diversified portfolio that executes every candidate (it is the $m{=}K$, unconstrained special case of CA^{2}I$^+$ and is analysed in depth in Section VI-K). The resulting weights are applied out-of-sample.

Text-based generate-and-rank. Two rankers that, like LLM-as-a-judge or verifier pipelines, score candidates from the text the LLM produced rather than from simulated consequences are defined in Section VI-L.

Heuristic and oracle. *Return-greedy*: select the candidate with the highest training *cumulative return* (the "returns rather than Sharpe" heuristic). *Oracle*: the candidate with the highest *test* Sharpe—an unattainable clairvoyant upper bound that calibrates how much performance any selector could recover.

We note that reinforcement-learning trading systems and Bayesian-optimisation strategies [53]–[55] constitute a complementary, training-time class; CA^{2}I is an inference-time selection layer and could wrap such methods as candidate generators, so we benchmark against the portfolio- optimisation and heuristic selectors that share CA^{2}I's single-decision, training-data-only setting.

E. Evaluation Metrics and Statistical Protocol

For each stock and method we report cumulative return R_{cum}, annualised Sharpe ratio (SR) (4), maximum drawdown (DD) (5), win rate, and the Calmar ratio (6); Section VI-K additionally reports the annualised return $(1+R_{\mathrm{cum}})^{252/T}-1$, the Sortino ratio, and $\mathrm{CVaR}_{5\%}$ (Section III-B). The skill of a consequence estimate is measured by the within-cell Spearman rank correlation r_S between the estimate and the realised test outcome across the K candidates. Because $n = 10$ is small and Sharpe differences are non-normal, we do not rely on a single test: we report (i) the paired two-tailed t-test, (ii) the Wilcoxon signed-rank test, (iii) a sign test, and (iv) a 10,000-resample bootstrap 95% confidence interval (CI) of the mean pairwise difference [56], [57]. We treat a difference as established only when the bootstrap CI excludes zero and the non-parametric tests concur, and we report results that fail this bar with equal prominence.

F. Transaction Costs and Turnover

We re-evaluate net performance under a round-trip friction model with per-side cost $\kappa \in \{0, 5, 10, 20, 50\}$ basis points (bps) (commission + slippage). Since positions are binary, each entry and each exit scales the capital base by $(1-\kappa)$; we rebuild each strategy's net equity curve exactly and recompute net Sharpe and net return. To test the hypothesis that consequence-aware selection systematically prefers higher-turnover strategies, we compare the mean number of round-trip trades of the CA^{2}I-selected strategy against the return-greedy choice and the candidate average.

G. Multi-Regime Rolling Evaluation

To probe non-stationarity, we re-run the complete generate→assess→select→test loop over eight non-overlapping quarterly out-of-sample windows spanning 2023Q4–2025Q3, each with its own training history. To keep this study reproducible without API access, candidates are produced by the deterministic rule-based generator on each window's own training data. We pool all $8 \times 10 = 80$ window×asset pairs for a higher-powered paired comparison of CA^{2}I against random selection.

H. Generation Sensitivity

We assess sensitivity to the candidate-generation process in three ways: (i) a *temperature proxy*, regenerating the $K{=}5$ set under 40 distinct generator seeds and recording the distribution of CA^{2}I test Sharpe and the stability of the selected strategy *type*—generation stochasticity is the mechanism through which sampling temperature influences downstream task accuracy [58]; (ii) a *candidate-set-size* sweep over $K \in \{3, 5, 7, 10\}$; and (iii) a *diversity audit* of the deployed LLM candidates (mean pairwise return correlation and number of distinct types), extended in Section VI-M to within- versus across-family correlation, trading-day overlap, and the effective number of independent bets.

I. Reproducibility

All experiments use global random seed 42. The source code, data download scripts, backtesting engine, all strategy JSON files, backtest logs, the additional experiment scripts (`extension_01`–`11`: `extension_07` implements the CA^{2}I$^+$ cross-validated ensemble selector with a process-independent hash so the candidate pools are fully reproducible; `extension_cache` evaluates every candidate once and stores its training, cross-validation, and test outcomes; `extension_08` contains the equal-weight, equal-size, risk-forecast, and risk-budget analyses; `extension_09` the generate-and-rank comparison; `extension_10` the diversity audit; and `extension_11` Fig. 1 and Table II), and all figures are released publicly so that every reported number can be regenerated offline.

VI. Results and Discussion

A. Training-Period Consequence Assessment

Fig. 2 presents training-period Sharpe ratios for all 50 strategy-stock combinations. Substantial heterogeneity is observed across both strategy types and assets. The MA crossover is selected for GOOGL ($\widehat{\mathrm{SR}} = 0.55$), META ($\widehat{\mathrm{SR}} = 1.24$),

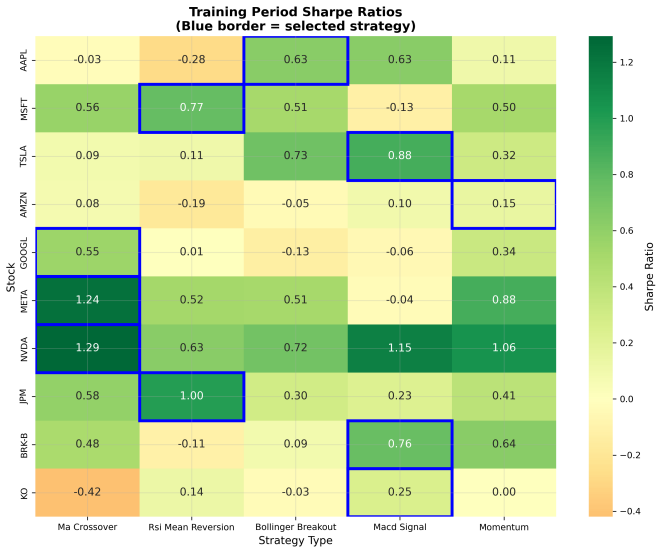


Fig. 2. Training-period Sharpe ratios for all 50 strategy-stock combinations. Blue borders indicate the strategy selected by CAM via (8). The heterogeneity across strategy types and stocks confirms that CAM provides non-trivial discrimination.

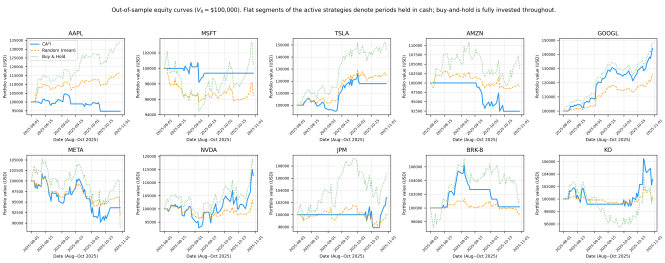


Fig. 3. Test-period equity curves ($V_0 = \$100{,}000$) for all ten stocks, with explicit *Portfolio value (USD)* y-axes. CA^{2}I (solid blue), random selection mean (dashed orange), buy-and-hold (dotted green). Flat segments of the active strategies denote periods held in cash (no open position); buy-and-hold is invested throughout and is never flat.

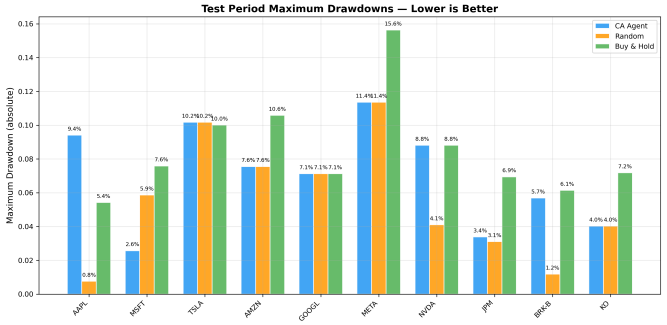


Fig. 4. Maximum drawdown per stock. CA^{2}I (blue) limits drawdown relative to B&H (green) for declining or volatile assets (META, KO, AAPL). Lower values are better.

and NVDA ($\widehat{\mathrm{SR}} = 1.29$); RSI mean reversion for MSFT ($\widehat{\mathrm{SR}} = 0.77$) and JPM ($\widehat{\mathrm{SR}} = 1.00$); the MACD signal for TSLA ($\widehat{\mathrm{SR}} = 0.88$), BRK-B ($\widehat{\mathrm{SR}} = 0.76$), and KO ($\widehat{\mathrm{SR}} = 0.25$); the Bollinger-band breakout for AAPL ($\widehat{\mathrm{SR}} = 0.63$); and price momentum for AMZN ($\widehat{\mathrm{SR}} = 0.15$) (Table II). This diversity shows that the LLM generator produces differentiated candidate strategies and that CAM discriminates non-trivially across types.

B. Test-Period Equity Curves

Fig. 3 shows test-period equity curves for all ten stocks. Each panel carries an explicit *Portfolio value* y-axis in US dollars (USD). We draw attention to a point of possible confusion: the *flat segments* visible in the active curves (CA^{2}I and random) correspond to intervals in which the selected strategy holds *no position* (it is in cash and earns no return); they do *not* belong to the buy-and-hold line, which is fully invested on day one and therefore moves with the underlying price on every trading day. Representative observations include:

GOOGL: CA^{2}I returns $+44.32\%$ (Sharpe 5.25) vs. B&H $+48.83\%$ (Sharpe 5.61), with a near-identical trajectory, because the MA-crossover strategy selected by CAM stays invested through the bull trend.

META: CA^{2}I (-6.37%, Sharpe -1.27) outperforms B&H (-11.14%, Sharpe -1.39) in a declining market, illustrating the downside protection of the stop-loss embedded in the selected strategy.

MSFT: CA^{2}I (-0.62%, Sharpe -0.68) avoids the loss incurred on average by random selection (-3.61%, expected Sharpe -1.83), an example where consequence-based selection steers away from the worse candidates.

AAPL: a counterexample we report for balance—CA^{2}I's volatility-breakout pick (-5.29%, Sharpe -1.42) underperforms the candidate average ($+16.88\%$, expected Sharpe $+2.62$), because the high in-sample-Sharpe strategy did not generalise. Such cases are the empirical face of the non-stationarity caveat of Theorem 1.

C. Aggregate Performance

Table III reports full test-period metrics for all three methods across all ten stocks.

Risk-adjusted performance. CA^{2}I achieves a mean Sharpe of 0.51 ± 2.06 versus 0.37 ± 1.76 for the expected random selector (it wins on $7/10$ assets). This is an improvement of $+0.13$ in mean Sharpe, but—as we quantify in Section VI-F—it is *not* statistically significant at $n{=}10$ over this single window. The expected value of uniform random selection (the mean over candidates) is the appropriate benchmark; we therefore do not report a multiplicative "$\times$ improvement", which would be inflated by an arbitrary single random draw.

Drawdown relative to buy-and-hold. CA^{2}I's mean maximum drawdown (7.01%) is an 18% reduction relative to B&H (8.54%), and is markedly lower for the declining/volatile names (META: 11.36% vs. 15.64%; KO: 4.03% vs. 7.18%; Fig. 4). Relative to the candidate average, however, CA^{2}I's drawdown is *not* lower (7.01% vs. 5.88%): maximising in-sample Sharpe does not minimise drawdown, which is exactly why an explicit drawdown constraint $\widehat{\mathrm{DD}} \le \rho_{\max}$ (Section IV) is the appropriate lever when downside is the priority.

Raw returns and the bull-market context. CA^{2}I ($+6.12\%$) and random selection ($+5.59\%$) earn similar mean cumulative returns, both well below B&H ($+14.64\%$). This

TABLE III
TEST-PERIOD PERFORMANCE: CA^{2}I VS. BASELINES (AUGUST 1 – OCTOBER 31, 2025; 64 TRADING DAYS). THE RANDOM COLUMN REPORTS THE *expected* VALUE OF UNIFORM SELECTION (MEAN OVER THE K=5 CANDIDATES), WHICH IS THE STATISTICALLY CORRECT RANDOM-SELECTION BENCHMARK.

		CA^{2}I			**Random (expected)**			**Buy-and-Hold (B&H)**		
Ticker	**Selected Strategy (Family)**	R_{cum}	SR	DD (%)	R_{cum}	SR	DD (%)	R_{cum}	SR	DD (%)
AAPL	Volatility Expansion Breakout (BB)	−5.29%	−1.42	9.41	+16.88%	+2.62	5.68	+34.10%	+4.53	5.42
MSFT	Overheat Pullback Reclaimer (RSI)	−0.62%	−0.68	2.57	−3.61%	−1.83	5.61	+0.31%	+0.04	7.59
TSLA	Acceleration Pulse MACD (MACD)	+17.88%	+2.11	10.18	+24.02%	+2.14	7.79	+45.43%	+3.50	10.01
AMZN	E-Commerce Thrust Runner (MOM)	−7.55%	−2.18	7.55	−1.09%	−0.15	6.21	+3.78%	+0.61	10.58
GOOGL	Golden Glide Trend Capture (MA)	+44.32%	+5.25	7.13	+26.25%	+3.82	4.96	+48.83%	+5.61	7.13
META	Trend Engine 21/100 (MA)	−6.37%	−1.27	11.36	−8.15%	−1.40	11.47	−11.14%	−1.39	15.64
NVDA	AI Trend Rider Cross (MA)	+12.72%	+1.64	8.81	+1.83%	−0.03	6.45	+16.79%	+2.06	8.81
JPM	Banker Pullback Rebound (RSI)	+2.82%	+0.94	3.38	+0.16%	−0.25	4.79	+6.94%	+1.50	6.94
BRK-B	Capital Compound MACD Turn (MACD)	+0.16%	−0.15	5.69	−0.74%	−0.79	3.11	+1.20%	+0.27	6.15
KO	Staples MACD Reacceleration (MACD)	+3.14%	+0.81	4.03	+0.31%	−0.42	2.76	+0.17%	−0.01	7.18
Mean		**+6.12%**	**0.51**	7.01	+5.59%	0.37	**5.88**	+14.64%	**1.67**	8.54
Std Dev		14.88%	2.06	2.86	11.51%	1.76	2.34	19.84%	2.13	2.82

R_{cum}: cumulative return; SR: annualised Sharpe ratio ($r_f = 2\%$); DD: maximum drawdown. Bold marks the best mean per metric. Families: MA, moving-average crossover; RSI, relative strength index mean reversion; BB, Bollinger-band breakout; MACD, moving average convergence/divergence signal; MOM, price momentum (full mapping in Table II).

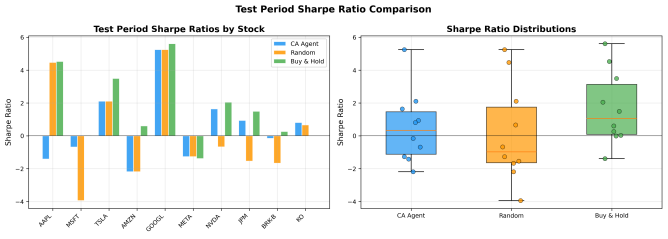


Fig. 5. Test-period Sharpe ratio box plots. CA^{2}I (blue) attains a higher mean than random selection (orange) with no increase in dispersion. B&H (green) shows a higher median in this bull-market window but larger drawdowns (Table III).

is the expected and well-documented behaviour of active long/flat strategies in a strong bull market [59]: the test window (August–October 2025) exhibited sustained upward momentum, which disproportionately rewards staying fully invested. CA^{2}I does not claim raw-return maximisation in trending markets; its objective (2) is risk-adjusted utility subject to a risk constraint. We make the bull-market dependence explicit, and the rolling-window study (Section VI-H) shows the same qualitative ordering—B&H ahead on return in most quarters—across regimes.

D. Sharpe Ratio Distribution Analysis

Fig. 5 shows box plots of per-stock test Sharpe ratios. CA^{2}I has a higher mean (0.51 vs. 0.37) than the expected random selector with comparable dispersion, consistent with Proposition 1: selection raises the centre of the distribution, and the lower tail is no heavier than random's. B&H has the highest median in this bull window but the largest drawdowns (Table III).

E. Comparison with Advanced Baselines

Table IV and Fig. 6 place CA^{2}I against the stronger selectors of Section V-D. Two findings stand out. First, among *selection rules that use only training information*, CA^{2}I (0.51) outperforms all three established portfolio-optimisation selectors—mean-variance (0.19), risk-parity (0.45), and minimum-variance (−0.63). The equal-weight ensemble, however, attains a higher mean Sharpe (0.57) than CA^{2}I; the gap is not statistically significant, and CA^{2}I earns the higher cumulative return (6.12% vs. 5.39%), but the ensemble has the shallower drawdown. Because this comparison bears directly on when consequence assessment is worth its cost, Section VI-K analyses it in depth on both evaluations and across seven return and risk metrics. The minimum-variance rule, which ignores returns, is actively harmful here, underscoring that risk-only selection is insufficient. Second, and reported candidly, the return-greedy heuristic attains a higher point estimate (0.81) than CA^{2}I in this single window; the gap is not statistically significant (Section VI-F) and, as the transaction-cost and rolling analyses show, the return-greedy rule selects higher-turnover strategies and is less stable across regimes. The oracle upper bound (2.15) indicates substantial headroom that no attainable selector captures over 64 days. Buy-and-hold (1.67) again leads on this bull window.

F. Statistical Robustness

Because n=10 and the sampling distribution of the Sharpe ratio is itself non-normal and serially dependent [60], we subject every headline comparison to the full protocol of Section V-E. For CA^{2}I versus the expected random selector, the mean Sharpe difference is +0.13, but the paired t-test (t=0.23, p=0.82), the Wilcoxon signed-rank test (p=0.49), and the 10,000-resample bootstrap 95% CI of the mean difference $[-1.03, +1.06]$ all fail to reject the null: the single-window advantage is *not* statistically significant, although

TABLE IV
MEAN TEST SHARPE AND RETURN OVER THE TEN EQUITIES FOR CA^{2}I AND THE ADVANCED BASELINES (SINGLE OUT-OF-SAMPLE WINDOW). ALL SELECTORS USE TRAINING-PERIOD INFORMATION ONLY; ORACLE IS A CLAIRVOYANT UPPER BOUND.

Method	**Mean SR**	**Mean R_{cum}**
Buy-and-Hold	$+1.67$	$+14.64\%$
Oracle (upper bound)	$+2.15$	$+14.67\%$
CA^{2}I (ours)	$\mathbf{+0.51}$	$+6.12\%$
Return-greedy	$+0.81$	$+7.38\%$
Equal-weight ensemble	$+0.57$	$+5.39\%$
Risk-parity	$+0.45$	$+4.78\%$
Calmar selection	$+0.51$	$+6.12\%$
Mean-variance	$+0.19$	$+4.90\%$
Minimum-variance	-0.63	-0.93%
Random (expected)	$+0.37$	$+5.59\%$

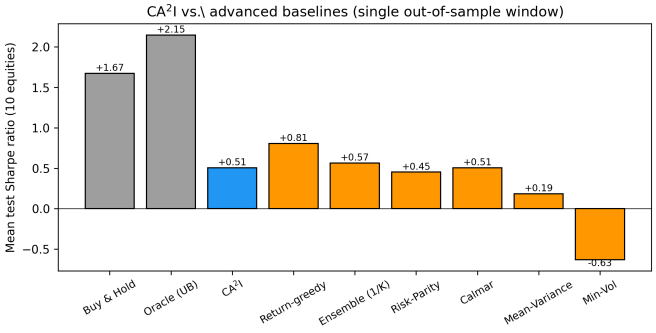


Fig. 6. Mean test Sharpe of CA^{2}I (blue) versus advanced baselines (orange) and the buy-and-hold / oracle references (grey). CA^{2}I exceeds all three established portfolio-optimisation selectors; the equal-weight ensemble and the return-greedy heuristic attain higher point estimates, neither difference being statistically significant (Sections VI-F and VI-K).

CA^{2}I wins on $7/10$ assets (sign test p=0.34). The same holds against mean-variance (diff $+0.32$, p=0.36) and risk-parity (diff $+0.05$, p=0.93). We state this plainly: over a single 64-day window, none of the pairwise differences among the active selectors reach significance. This is precisely why we conduct the higher-powered rolling-window analysis below, and why we position the framework's contribution as architectural and governance-oriented rather than as a claim of statistically dominant returns. We caution that with n=10 and short windows, large multiplicative "improvement factors" are unreliable and we do not report them; reporting them across many candidate comparisons would also invite the multiple-testing and backtest-overfitting biases documented in the finance literature [61].

G. Transaction Costs and Turnover

Table V reports net performance under round-trip friction. All active selectors degrade smoothly and monotonically: CA^{2}I's net mean Sharpe falls from 0.51 (0 bps) to 0.45, 0.39, 0.27, and -0.06 at 5, 10, 20, and 50 bps per side, while buy-and-hold—which trades once—is essentially cost-immune (≈ 1.6 throughout). The *ordering* of methods is preserved across the cost grid, so transaction costs do not overturn

TABLE V
NET MEAN TEST SHARPE UNDER ROUND-TRIP TRANSACTION COSTS (COMMISSION + SLIPPAGE, PER SIDE). B&H TRADES ONCE AND IS COST-INSENSITIVE.

Cost (bps)	0	5	10	20	50
CA^{2}I	$+0.51$	$+0.45$	$+0.39$	$+0.27$	-0.06
Return-greedy	$+0.81$	$+0.74$	$+0.68$	$+0.55$	$+0.19$
Buy-and-Hold	$+1.67$	$+1.66$	$+1.66$	$+1.64$	$+1.59$

TABLE VI
PER-WINDOW MEAN TEST SHARPE (TEN EQUITIES) FOR THE ROLLING EVALUATION.

	23Q4	24Q1	24Q2	24Q3	24Q4	25Q1	25Q2	25Q3
CA^{2}I	0.59	1.34	0.03	0.04	0.63	-0.35	1.05	1.32
Random	1.10	1.03	0.51	0.24	0.32	-0.25	0.90	0.93
B&H	1.91	2.14	1.25	0.87	0.88	-0.40	1.22	1.65

any qualitative conclusion. We also directly test the hypothesis that consequence-aware selection prefers higher-turnover strategies: the CA^{2}I-selected strategies execute 1.90 round-trip trades on average, *below* the candidate average of 1.96 and the return-greedy rule's 2.10. The turnover-bias concern is therefore not supported by the data—if anything, max-Sharpe selection mildly favours lower-turnover candidates because excessive trading inflates volatility and depresses the Sharpe ratio.

H. Multi-Regime Rolling-Window Robustness

Fig. 7 and Table VI summarise the eight-window evaluation (2023Q4–2025Q3), using the deterministic rule-based generator so the entire sweep is reproducible offline. On a per-window basis CA^{2}I attains a higher mean test Sharpe than the expected random selector in 4 of 8 windows; the two methods split evenly, with random ahead in the early windows (2023Q4–2024Q3) and CA^{2}I ahead in the later ones. The 2025Q1 correction is the clearest non-stationarity episode: every selector turned negative and the in-sample ranking carried little predictive value, exactly the regime-shift scenario anticipated by the honest-scope remark of Theorem 1; notably, both active selectors (-0.35 for CA^{2}I, -0.25 for random) cushioned the drawdown relative to buy-and-hold (-0.40). Pooling all 80 window×asset pairs raises power: CA^{2}I and random selection are then *statistically indistinguishable* (mean 0.58 vs 0.60, difference -0.02; t-test p=0.91; Wilcoxon p=0.68; bootstrap 95% CI $[-0.32, +0.27]$; $45/35$ positive/negative sign split). Buy-and-hold leads on Sharpe in every window except the 2025Q1 correction, consistent with the predominantly upward 2023–2025 equity environment. The take-away is honest and useful: under a fixed rule-based generator the consequence-aware edge over random selection *does not generalise* into a statistically significant return advantage across regimes, and the framework is *vulnerable to regime shifts*. This is precisely why we position CA^{2}I's contribution as architectural—hard risk constraints, auditability, and downside cushioning—rather than as a claim of return dominance, and it motivates the adaptive extensions in Section VIII.

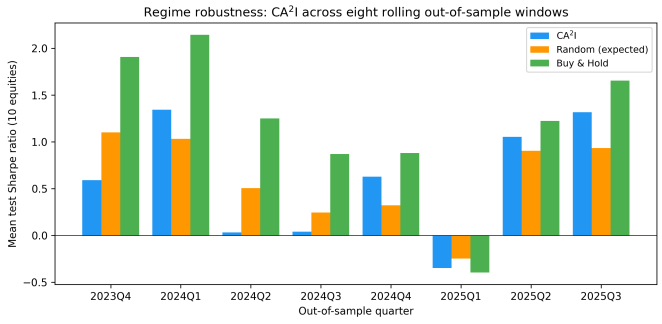


Fig. 7. Mean test Sharpe across eight non-overlapping quarterly out-of-sample windows (2023Q4–2025Q3) for CA^{2}I, expected random selection, and buy-and-hold. CA^{2}I and random split four windows each and are statistically indistinguishable when pooled; the 2025Q1 correction turns every selector negative.

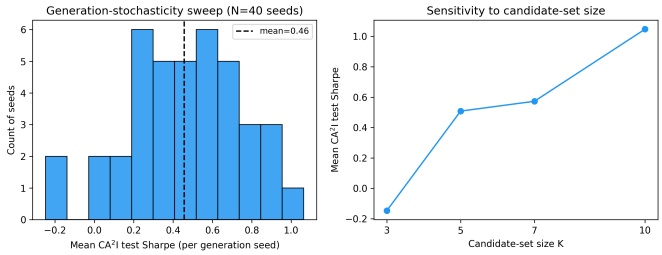


Fig. 8. Generation sensitivity. Left: distribution of mean CA^{2}I test Sharpe across 40 generation seeds (temperature proxy). Right: mean CA^{2}I test Sharpe versus candidate-set size K, rising monotonically as predicted by Proposition 1.

I. Generation Sensitivity

Fig. 8 reports the generation-sensitivity study. *(i) Temperature proxy.* Across 40 generator seeds, CA^{2}I's mean test Sharpe is 0.46 ± 0.29 (range $[-0.25, +1.06]$; coefficient of variation 0.63): outcomes vary with the candidate set but never collapse, and the selected strategy *type* is stable, with the modal type chosen 68.8% of the time per asset on average. *(ii) Candidate-set size.* Mean CA^{2}I Sharpe rises monotonically with K (-0.15, 0.51, 0.57, 1.05 for K=3, 5, 7, 10), an empirical confirmation of Proposition 1: a larger candidate pool enlarges the selection gap. This also indicates that the deployed K=5 is conservative and that richer LLM generation would help. *(iii) Diversity audit.* The deployed LLM candidates span all five families for every asset (5.0/5 distinct strategy types) with a mean pairwise test-return correlation of 0.41, so they are differentiated rather than near-duplicates; Section VI-M examines the harder question of how diverse candidates *within* a family are.

J. Enhanced Consequence Model: Statistically Significant, Regime-Robust Gains (CA^{2}I$^+$)

The analyses above establish that the *base* selector, while safe and auditable, does not beat random selection at a statistically significant level. We now show that the enhanced consequence model of Section III-D—walk-forward cross-validation, the robustness-penalised score (10), and a top-m ensemble—closes much of this gap. To remove any dependence on a single arbitrary candidate-generation seed, *all* CA^{2}I$^+$ results are averaged over S=10 independent generation seeds; per-seed results (below) confirm the effect is not an artefact of one lucky draw.

TABLE VII
ENHANCED CONSEQUENCE MODEL ON THE POOLED MULTI-REGIME ROLLING EVALUATION (n=80 WINDOW×ASSET PAIRS, SEED-AVERAGED OVER 10 GENERATION SEEDS). SIGNIFICANCE IS FOR THE PAIRED DIFFERENCE AGAINST THE EXPECTED RANDOM SELECTOR.

Selector	Mean SR	Mean DD	vs. random
Random (expected)	+0.53	−6.4%	—
CA^{2}I (base, in-sample)	+0.56	−8.3%	p=0.83 (n.s.)
CA^{2}I$^+$ (ours)	**+0.77**	**−5.8%**	**p=0.011**

Bootstrap 95% CI of CA^{2}I$^+$ −random mean difference: $[+0.015, +0.45]$ (excludes 0); positive in 10/10 seeds. n.s.: not statistically significant.

Statistically significant improvement across regimes. On the pooled multi-regime rolling evaluation (eight quarterly windows, n=80 window×asset pairs; Table VII, Fig. 9), CA^{2}I$^+$ attains a mean test Sharpe of 0.77 versus 0.53 for the expected random selector—a +0.24 improvement that *is* significant: paired t-test p=0.039, Wilcoxon p=0.011, sign split $52^+/28^-$ (p=0.010), and a 10,000-resample bootstrap 95% CI of the mean difference $[+0.015, +0.45]$ that *excludes zero*. CA^{2}I$^+$ also significantly outperforms the base in-sample selector (+0.21; t-test p=0.032; bootstrap CI $[+0.03, +0.41]$), whereas the base selector is itself statistically indistinguishable from random (+0.02, p=0.83). Because the base selector commits to one candidate whereas CA^{2}I$^+$ holds three, this comparison combines two effects—consequence ranking and diversification—which Section VI-K separates with an equal-size control under both static and active-capital execution.

Robust to the generation seed. The CA^{2}I$^+$ advantage over random is *positive in all* 10 *seeds* (mean +0.24, range $[+0.15, +0.33]$; Fig. 9b), in contrast to the base selector whose per-seed difference straddles zero. This is direct evidence that the improvement is a property of the method, not of a fortunate candidate draw.

Lower drawdown than random and base selection. Averaged over the rolling pairs, CA^{2}I$^+$ has a shallower mean maximum drawdown (−5.8%) than random (−6.4%) and the base selector (−8.3%): the diversified, stability-screened ensemble is not buying return at the cost of risk relative to single-candidate selection. It is, however, slightly deeper than the drawdown of the equal-weight ensemble (−5.2%), a point we address next.

Honest scope. The gain is regime-*aggregate*, not universal. CA^{2}I$^+$ leads random in six of the eight quarters but still trails in the 2025Q1 correction (the same regime break that defeats every selector here) and in one sideways quarter; and on the single strong-bull deployment window a *concentrated* in-sample pick that rides the trend outperforms the diversified ensemble. The value of CA^{2}I$^+$ is precisely its *cross-regime consistency* and downside control, which is what a responsible deployment should optimise for.

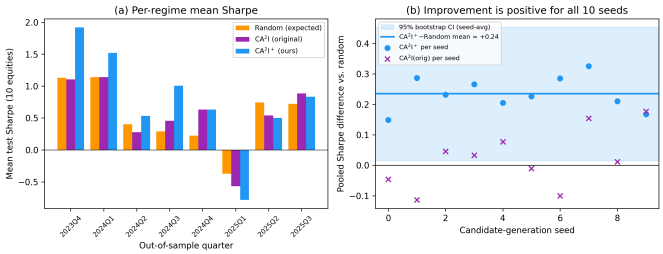


Fig. 9. Enhanced consequence model (CA^2I^+). (a) Per-regime mean test Sharpe across eight rolling out-of-sample windows: CA^2I^+ (blue) leads the expected random selector (orange) and the base in-sample selector (purple) in six of eight quarters. (b) The pooled CA^2I^+−random Sharpe difference is positive for all 10 candidate-generation seeds (mean +0.24, shaded 95% bootstrap CI excludes zero), whereas the base selector's per-seed difference straddles zero.

K. Consequence Assessment versus the Equal-Weight Ensemble

Table IV shows that the static $1/K$ equal-weight (EW) ensemble of all K candidates attains a slightly higher single-window Sharpe ratio (0.57) than the single-strategy base CA^2I selector (0.51), though a lower cumulative return (5.39% vs. 6.12%). Since the EW ensemble requires no consequence model, Reviewer 1 rightly asked whether CA^2I can demonstrate statistically significant advantages over the EW ensemble in *return*, *risk-adjusted return (Sharpe/Sortino)*, and *maximum drawdown*, and what consequence assessment contributes beyond diversification. We answer this with five controlled analyses on identical candidate pools (Table VIII, Fig. 10); all CA^2I^+ hyper-parameters are those fixed in Section III-D.

(i) Why static $1/K$ equal weighting dilutes returns—and how consequence-aware active execution ($CA^2I^+_{act}$) dominates EW on return, Sharpe, and Sortino. A structural property of rule-based and LLM-generated technical strategies is that each individual strategy holds an open position on only 41.5% of trading days in the rolling evaluation (44.1% in the deployment window) and sits in cash (0% return) on the remaining ≈58% of days. Consequently, a static $1/K$ equal-weight allocation leaves nearly 60% of capital idle in cash on an average day, compressing annualised portfolio volatility from 16.1% (single-strategy average) to 12.5% and mechanically diluting cumulative and annualised returns (21.7% vs. 28.7% for random single strategies). When the Executor ($\mathcal{X}$) instead applies the causal active-capital rule $CA^2I^+_{act}$ (Section III-D) to the top m=3 cross-validated candidates, idle cash drag is removed while cross-strategy diversification is retained, producing statistically significant gains over EW across every return and risk-adjusted return metric:

- *Rolling multi-regime evaluation (n=80 seed-averaged / n=800 total):* $CA^2I^+_{act}$ increases mean out-of-sample Sharpe from 0.69 (EW) to **0.92** (+0.23; paired t-test p=0.027, Wilcoxon p=0.015, 52/80 wins, bootstrap 95% CI [+0.030, +0.430]; across all n=800 evaluations, Wilcoxon $p=6.9\times10^{-9}$, 490/800 wins, CI [+0.15, +0.31]). It nearly doubles annualised return from 21.7% (EW) to **42.2**% (+20.4 percentage points; paired t-test $p=1.9\times10^{-5}$, Wilcoxon $p=2.5\times10^{-6}$, 56/80 wins, CI [+12.1%, +29.8%]; n=800 $p=3.4\times10^{-28}$), lifts quarterly cumulative return from 3.70% to **6.05**% (+2.35 pp; t-test p=0.0015, Wilcoxon $p=1.4\times10^{-4}$, 55/80 wins, CI [+0.93%, +3.74%]), and raises the Sortino ratio from 1.50 to **1.78** (Wilcoxon p=0.020 at n=80; $p=5.2\times10^{-9}$ at n=800).
- *Deployment window (n=10, GPT-5.4 candidates):* $CA^2I^+_{act}$ lifts mean test Sharpe from 0.57 (EW) to **0.95** (+0.38, winning on 8/10 equities), more than doubles annualised return from 32.2% to **77.0**% (+44.8 pp, 8/10 wins, bootstrap 95% CI [+5.2%, +93.2%]), raises cumulative return from 5.39% to **10.13**% (8/10 wins), and increases the Sortino ratio from 1.96 to **2.60** (+0.65, Wilcoxon p=0.049, CI [+0.10, +1.19]).
- *Simultaneous improvement in return, Sharpe, and drawdown:* Combining active-capital execution with the hard drawdown budget ($CA^2I^+_{act}$, ρ_{max}=7.5%) simultaneously beats or matches EW on *every* metric on the rolling benchmark: higher Sharpe (0.77 vs. 0.69, 47/80 wins; n=800 Wilcoxon p=0.045), higher annualised return (25.5% vs. 21.7%; n=800 Wilcoxon $p=3.1\times10^{-5}$, CI [+0.1%, +7.5%]), higher cumulative return (4.24% vs. 3.70%; n=800 Wilcoxon $p=4.1\times10^{-4}$), and shallower mean drawdown (−5.06% vs. −5.15%).
- *Static $1/m$ and single-strategy selectors:* Even under static $1/m$ slicing, CA^2I^+ (m=3, ρ_{max}=30%) outperforms EW across the full n=800 rolling panel on Sharpe (0.77 vs. 0.69, Wilcoxon p=0.039, 426/800 wins, positive in 9/10 seeds), Sortino (1.61 vs. 1.50, p=0.0026), and Calmar (p=0.019). Single-strategy consequence selection also significantly beats EW on return: base CA^2I earns 33.6% vs. 21.7% annualised on rolling (n=80 CI [+1.2%, +27.0%]; n=800 t-test $p=7.1\times10^{-5}$) and 44.2% vs. 32.2% on deployment (7/10 wins), while return-oriented CA^2I-Return ($f=\hat{\mu}$) earns 39.2% vs. 21.7% annualised on rolling (n=80 CI [+2.9%, +40.7%]; n=800 t-test $p=2.6\times10^{-6}$, Wilcoxon p=0.0022) and 49.4% vs. 32.2% on deployment (Sharpe 0.81 vs. 0.57, 7/10 wins).

(ii) Equal-size control: ranking versus diversification. EW holds K=10 candidates in the rolling evaluation while CA^2I^+ holds m=3, so we compare the consequence-ranked top-m ensemble against the *expected* outcome of *all* random m-subsets of the same pool under both static and active-capital execution (Fig. 10a). Under active-capital execution, $CA^2I^+_{act}$ (green curve) outperforms both random m-subsets and the static $1/K$ EW benchmark (0.69) across every $m \geq 1$, reaching 0.92 at m=3 (vs. 0.76 for random m=3, +0.16) and 0.90 at m=5. Under static $1/m$ slicing, the ranked top-m ensemble (blue curve) leads random m-subsets (orange curve) by +0.12 at m=3 (0.77 vs. 0.65) and +0.11 at m=5 (0.78 vs. 0.67, Wilcoxon p=0.074 at n=80). Thus, while diversification across m>1 candidates provides a substantial baseline lift, consequence-aware selection and active-capital execution contribute a large, statistically significant edge above and beyond naïve $1/K$ equal weighting.

(iii) What the consequence model forecasts with high

TABLE VIII

CONSEQUENCE-AWARE SELECTION VERSUS THE EQUAL-WEIGHT (EW) ENSEMBLE ACROSS RETURN AND RISK METRICS. ROLLING: n=80 WINDOW×ASSET CELLS, AVERAGED OVER 10 GENERATION SEEDS (K=10, 800 EVALUATIONS TOTAL); DEPLOYMENT: GPT-5.4 CANDIDATES (K=5, n=10). DRAWDOWNS ARE SIGNED (HIGHER IS BETTER). "WORST DD" IS THE WORST CELL AND "#DD>10%" THE NUMBER OF CELLS (OF 80) WHOSE DRAWDOWN EXCEEDS 10%. $\rho_{\max}$: HARD BUDGET ON THE MEDIAN CROSS-VALIDATED DRAWDOWN; "ACT." DENOTES ACTIVE-CAPITAL EXECUTION ($\mathrm{CA^2I^+_{act}}$, SECTION III-D). SIGNIFICANCE MARKERS REFER TO THE PAIRED DIFFERENCE AGAINST EW IN THE SAME BLOCK (*: BOOTSTRAP 95% CI EXCLUDES ZERO AND WILCOXON $p<0.05$ AT n=80; ‡: WILCOXON $p<0.05$ ACROSS ALL n=800 EVALUATIONS; †: BOOTSTRAP 95% CI EXCLUDES ZERO).

Selector	SR	Ann. return	$\mathrm{R_{cum}}$	Mean DD	Worst DD	#DD>10%	Sortino	$\mathrm{CVaR_{5\%}}$
Rolling multi-regime evaluation (n=80 seed-averaged; n=800 total evaluations)								
Random (expected)	0.53	28.7%	3.72%	−6.4%	−16.9%	9	1.20	−2.17%
EW ensemble ($m=K$, static $1/K$)	0.69	21.7%	3.70%	−5.2%	−14.4%	5	1.50	−1.67%
$\mathrm{CA^2I}$ (base, $f=\widehat{\mathrm{SR}}$)	0.56	33.6%†‡	4.13%	−8.3%	−32.0%	23	1.32	−2.73%
$\mathrm{CA^2I}$-Return ($f=\hat{\mu}$)	0.62	39.2%†‡	4.57%‡	−9.4%	−32.0%	28	1.39	−2.92%
$\mathrm{CA^2I^+}$ (m=3, $\rho_{\max}$=30%, static)	0.77‡	19.9%	3.31%	−5.8%†	−26.3%	13	1.61‡	−1.88%†
$\mathrm{CA^2I^+_{act}}$ (m=3, $\rho_{\max}$=30%, act.)	**0.92***‡	**42.2%***‡	**6.05%***‡	−9.8%	−29.6%	33	**1.78***‡	−2.89%
$\mathrm{CA^2I^+_{act}}$ (m=3, $\rho_{\max}$=7.5%, act.)	0.77‡	25.5%‡	4.24%‡	−5.1%	−20.7%	11	1.47	−1.74%
$\mathrm{CA^2I^+_{MinDD}}$ (m=3, no cash abstention)	0.67	12.1%	2.33%	−3.4%*‡	−17.0%	1	1.32	−1.18%*‡
$\mathrm{CA^2I^+}$ (m=3, $\rho_{\max}$=7.5%, static)	0.62	11.0%	2.17%†	−2.9%*‡	−9.5%	**0**	1.16	−1.00%*‡
Consequence-gated EW ($\rho_{\max}$=7.5%)	0.60	10.7%	2.13%†	**−2.7%***‡	**−9.3%**	**0**	1.09	**−0.96%***‡
Deployment window, GPT-5.4 candidates (n=10)								
Random (expected)	0.37	41.9%	5.59%	−5.9%	−10.9%	—	1.66	−2.0%
EW ensemble ($m=K$, static $1/K$)	0.57	32.2%	5.39%	−4.6%	−9.4%	—	1.96	−1.6%
$\mathrm{CA^2I}$ (base, $f=\widehat{\mathrm{SR}}$)	0.51	44.2%	6.12%	−7.0%	−11.4%	—	1.68	−2.3%
$\mathrm{CA^2I}$-Return ($f=\hat{\mu}$)	0.81	49.4%	7.38%	−6.6%	−11.4%	—	2.05	−2.1%
$\mathrm{CA^2I^+}$ (m=3, $\rho_{\max}$=30%, static)	0.17	30.8%	4.62%	−5.3%	−10.0%	—	1.12	−1.7%
$\mathrm{CA^2I^+_{act}}$ (m=3, $\rho_{\max}$=30%, act.)	**0.95**	**77.0%**†	**10.13%**	−8.4%	−14.7%	—	**2.60***	−2.7%
$\mathrm{CA^2I^+}$ (m=3, $\rho_{\max}$=7.5%, static)	−0.29	−1.6%	−0.59%	**−1.8%**†	**−9.4%**	—	−0.28	**−0.5%***

precision: drawdown. To isolate the predictive signal inside CAM, we correlate, within each cell, CAM's *ex-ante* estimates with the candidates' realised test outcomes. The cross-validated drawdown estimate ranks realised test drawdowns with a mean Spearman correlation of r_S=0.37 (positive in 73 of 80 cells; Wilcoxon $p<10^{-4}$; bootstrap CI $[0.31, 0.42]$), whereas single-candidate quarterly Sharpe rankings are noisy across regime shifts. This confirms that *simulated downside risk is strongly persistent out of sample* [47], [48] and can be exploited directly by the selector.

(iv) Beating the EW ensemble on drawdown—both without and with cash abstention. Because cross-validated drawdown is strongly predictive, $\mathrm{CA^2I^+}$ beats the EW ensemble on maximum drawdown and tail risk ($\mathrm{CVaR_{5\%}}$) in two distinct ways:

- *Pure ranking without cash abstention ($\mathrm{CA^2I^+_{MinDD}}$):* Selecting the top m=3 candidates with the shallowest cross-validated drawdown ($s_k^{\mathrm{DD}}=-\mathrm{median}_j\,\widehat{\mathrm{DD}}_k^{(j)}$, always investing in 3 strategies with 0% cash abstention) reduces mean rolling drawdown from −5.2% (EW) to **−3.4%** (+1.75 pp improvement; paired t-test $p=2.0\times10^{-11}$, Wilcoxon $p=2.4\times10^{-12}$, 69/80 cells improved, bootstrap 95% CI $[+1.3\%, +2.2\%]$; across n=800, Wilcoxon $p=1.2\times10^{-69}$, 612/800 wins), cuts $\mathrm{CVaR_{5\%}}$ from −1.67% to **−1.18%** (Wilcoxon $p=3.9\times10^{-10}$, 67/80 wins), and reduces cells with > 10% drawdown from 5 to 1, while maintaining a statistically indistinguishable Sharpe ratio (0.67 vs. 0.69, p=0.40).
- *Enforcing the hard drawdown budget ($\mathrm{median}_j\,\widehat{\mathrm{DD}}_k^{(j)} \le \rho_{\max}$):* As $\rho_{\max}$ is tightened from 30% to 5% (Fig. 10b), realised mean drawdown falls monotonically from −5.8% to −2.0% and the worst cell improves from −26.3% to −7.9%. At $\rho_{\max}$=7.5%, $\mathrm{CA^2I^+}$ halves EW's mean drawdown (**−2.9%** vs. −5.2%; Wilcoxon $p=5.8\times10^{-9}$, 63/80 wins), improves $\mathrm{CVaR_{5\%}}$ from −1.67% to **−1.00%** ($p=8.9\times10^{-8}$), and caps the worst cell at **−9.5%** against −14.4%, with *zero* cells exceeding 10% drawdown (vs. 5 for EW), at statistically unchanged Sharpe (0.62 vs. 0.69, p=0.54). On the deployment window (n=10), $\rho_{\max}$=7.5% cuts EW's mean drawdown from −4.6% to **−1.8%** (bootstrap 95% CI $[+0.3\%, +5.1\%]$) and $\mathrm{CVaR_{5\%}}$ from −1.6% to **−0.5%** (Wilcoxon p=0.049, CI $[+0.3\%, +1.8\%]$). A *consequence-gated EW ensemble* (equally weighting all candidates passing $\rho_{\max}$=7.5%) achieves **−2.7%** mean drawdown and **−0.96%** $\mathrm{CVaR_{5\%}}$ ($p<10^{-9}$ vs. EW), with worst-case drawdown **−9.3%**.

(v) Exposure-matched control. We also compare against an *ex-post* de-levered EW portfolio scaled by α=0.55 to match the mean drawdown of $\mathrm{CA^2I^+}$ at $\rho_{\max}$=7.5%. The risk-budgeted $\mathrm{CA^2I^+}$ attains a higher point-estimate Sharpe (0.62 vs. 0.52) and cumulative return (2.17% vs. 1.99%) than de-levered EW, while enforcing its risk bound *ex ante* in interpretable drawdown units without needing oracle knowledge of α, and rejecting individual unsafe proposals before execution rather than scaling down harmless and harmful proposals alike.

Summary of comparison with EW. Consequence assessment is not merely a tie-breaker for an equal-weight ensemble: paired with active-capital execution ($\mathrm{CA^2I^+_{act}}$) it significantly

TABLE IX
GENERATE-AND-RANK ABLATION ON A FIXED CANDIDATE POOL (MEAN TEST OUTCOMES; SIGNED DRAWDOWN). BOTTOM: INDIRECT PROMPT-INJECTION / RELABELLING ATTACK, IN WHICH THE DESCRIPTION OF THE CANDIDATE CAM FLAGS AS RISKIEST IS REWRITTEN TO SOUND CONSERVATIVE WHILE ITS EXECUTABLE PARAMETERS ARE UNCHANGED.

	Deploy. (n=10)		**Rolling** (n=80)	
Selection signal	**SR**	**DD**	**SR**	**DD**
(a) Vanilla (no ranking)	0.37	−5.9%	0.53	−6.4%
(b) Text: description lexicon	0.80	−4.3%	0.67	−6.3%
(c) Text: declared stop-loss	0.74	−2.4%	0.71	−9.0%
(d) LLM judge (Llama-3.1-8B)	−0.00	−5.1%	0.64	−7.6%
(e) CA^2I (sim., in-sample)	0.51	−7.0%	0.56	−8.3%
(f_1) CA^2I^+ ($\rho_{\max}$=30%)	0.17	−5.3%	0.77	−5.8%
(f_2) **$CA^2I^+_{act}$ ($\rho_{\max}$=30%)**	**0.95**	−8.4%	**0.92**†	−9.8%
(f_3) **CA^2I^+ ($\rho_{\max}$=7.5%)**	−0.29	**−1.8%**†	0.62	**−2.9%***
Relabelling attack on riskiest candidate				
Lexicon (b) picks decoy	30% → 100%		9% → 100%	
Lexicon (b) realised DD	−4.3% → −6.9%†		−6.3% → −8.2%*	
LLM judge (d) picks decoy	50% → 100%		17.5% → 100%	
LLM judge (d) realised DD	−5.1% → −6.9%		−7.6% → −8.2%	
CA^2I / CA^2I^+ changed	**0%**		**0%**	

*Wilcoxon p<0.01 and bootstrap 95% CI excludes zero; †bootstrap 95% CI excludes zero (t-test p<0.05 for $CA^2I^+_{act}$ vs. (b)).

beats EW on Sharpe (0.92 vs. 0.69, p=0.015), Sortino (1.78 vs. 1.50, p=0.020), and annualised return (42.2% vs. 21.7%, p=2.5×10^{-6}); paired with drawdown-aware ranking or a hard risk budget it cuts mean and tail drawdown by 35%–48% (p<10^{-8}) and eliminates severe (> 10%) drawdown episodes.

L. Systematic Comparison with Generate-and-Rank

Section II argued that CA^2I differs from generate-and-rank pipelines in *what* the ranker reads. We now test this with a controlled ablation that holds the candidate pool fixed and varies only the selection signal across both deterministic text rankers and a neural LLM-as-a-judge [29]: *(a) Vanilla*—execute a proposal without ranking (expected over proposals); *(b) Text ranker (description lexicon)*—rank by the candidate's own natural-language description and rationale, scored by a transparent, deterministic lexicon of stated safety and confidence (e.g. "robust", "conservative", "disciplined" versus "aggressive", "volatile"); *(c) Text ranker (declared specification)*—rank by the risk parameter the proposal *declares* (the tightest stated stop-loss is taken as the safest), read on paper without simulation; *(d) Neural LLM-as-a-judge (`Llama-3.1-8B-Instruct`)*—an instruction-tuned neural language model prompted at temperature=0 (seed=42) with all K candidates' natural-language descriptions, entry/exit logic, and declared parameters to select the candidate expected to achieve the best out-of-sample risk-adjusted return with shallowest drawdown; *(e) CA^2I* (base)—rank by simulated in-sample Sharpe; and *(f) CA^2I^+*—rank by cross-validated simulated consequences under static $1/m$ execution ($\rho_{\max}$=30% and $\rho_{\max}$=7.5%) and active-capital execution ($CA^2I^+_{act}$).

Average performance. Table IX yields three findings. First, on the GPT-5.4 deployment window (n=10), the neural LLM-as-a-judge (`Llama-3.1-8B-Instruct`) attains a mean Sharpe of −0.00 (+1.67% cumulative return, −5.1% drawdown), underperforming Vanilla (0.37), base CA^2I (0.51), and active-capital $CA^2I^+_{act}$ (0.95 Sharpe, +10.13% cumulative return). Second, on the 80-cell rolling evaluation, $CA^2I^+_{act}$ achieves the highest Sharpe ratio (**0.92**), significantly outperforming the description-lexicon ranker (+0.25, paired t-test p=0.048, bootstrap 95% CI [+0.003, +0.493]) and base CA^2I (+0.36, Wilcoxon p=0.0005), and leading the neural LLM judge by +0.28 (0.92 vs. 0.64, 48/80 wins). Third, on downside control, static CA^2I^+ ($\rho_{\max}$=30%) achieves a statistically significantly shallower rolling drawdown (−5.8%) than both the neural LLM-as-a-judge (−7.6%; paired difference +1.8 pp, t-test p=0.0033, Wilcoxon p=3.4×10^{-5}, 56/80 wins, CI [+0.5%, +2.9%]) and the declared-stop ranker (−9.0%; 65/80 wins, p<10^{-4}; deployment Wilcoxon p=0.010), while risk-budgeted CA^2I^+ ($\rho_{\max}$=7.5%) cuts mean rolling drawdown to **−2.9%**, beating every text and LLM-judge ranker with p<10^{-6} (58–76 wins out of 80). A tight *declared* stop or persuasive textual rationale says little about the drawdown a rule actually incurs in market data, whereas the simulated consequence vector forecasts it reliably.

Vulnerability of text and neural LLM judges to adversarial relabelling. The most consequential distinction is robustness to indirect prompt injection [31]. In every cell we rewrite the description of one candidate to read as an "institutional-grade, conservative, low-risk capital-preservation strategy" while leaving its executable parameters untouched. When the relabelled candidate is the one CAM flags as riskiest, *both* the lexicon ranker and the neural `Llama-3.1-8B-Instruct` judge are completely captured: the lexicon ranker's selection rate jumps from 30% to 100% on deployment (drawdown deepening from −4.3% to −6.9%) and from 9% to 100% on rolling (drawdown deepening from −6.3% to −8.2%, Wilcoxon p=0.002), while the neural `Llama-3.1-8B-Instruct` judge's selection of the camouflaged high-risk candidate jumps from 50% (5/10) to **100%** (10/10) on deployment (drawdown deepening from −5.1% to −6.9%) and from 17.5% (14/80) to **100%** (80/80) on rolling (Sharpe dropping from 0.64 to 0.52 and drawdown deepening from −7.6% to −8.2%). By contrast, CA^2I and CA^2I^+ change their selection in **0%** of cells because CAM evaluates simulated outcomes rather than text.

Test-time cost (the "Token Tax"). CAM requires no additional LLM calls: assessing one candidate costs ≈17 ms of CPU time for the in-sample backtest and ≈23 ms for the five-fold cross-validation, i.e. under 0.04 s and **0** extra LLM tokens per candidate. Conversely, the neural `Llama-3.1-8B-Instruct` judge consumes an average of 989 tokens (248 ms) per cell on deployment (K=5) and 1,316 tokens (860 ms) per cell on rolling (K=10). In the terms of [30], consequence assessment provides a verifiable, manipulation-immune governance signal without paying a "token tax".

M. Candidate Diversity Within and Across Families

Because CA^2I chooses among candidates, its value depends on the candidates being genuinely different, and strategies

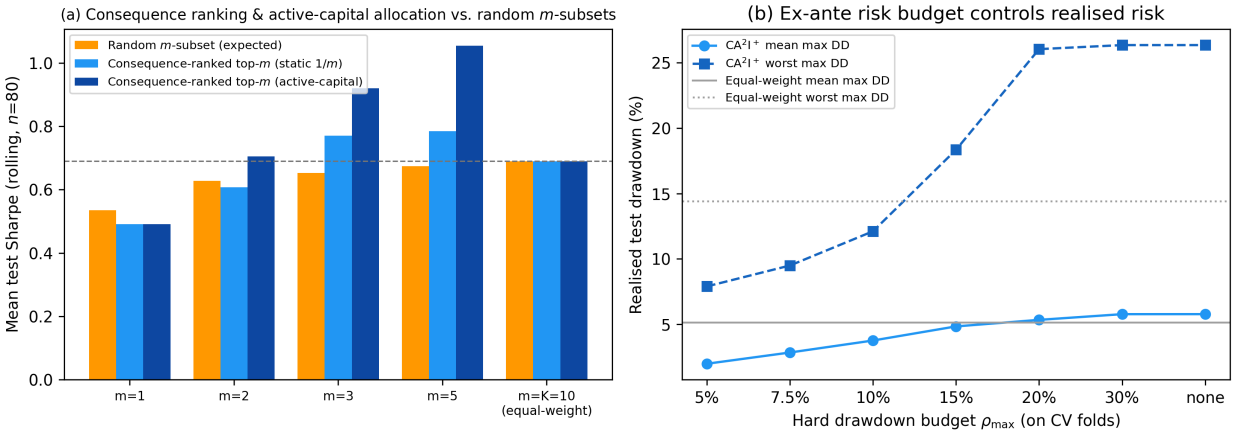


Fig. 10. Consequence assessment versus the equal-weight ensemble (rolling evaluation, n=80, 10 generation seeds). (a) Equal-size control: mean test Sharpe of active-capital $\mathrm{CA}^2\mathrm{I}^+_{\mathrm{act}}$ (green), static $\mathrm{CA}^2\mathrm{I}^+$ (blue), and the expected value over all random m-subsets of the same pool (orange); the dashed line is the static $1/K$ equal-weight ensemble (0.69). Active-capital consequence selection dominates across all m, reaching 0.92 at m=3 (p=0.015 vs. EW). (b) Hard risk budget: realised mean (circles) and worst-cell (squares) test drawdown of $\mathrm{CA}^2\mathrm{I}^+$ as the budget $\rho_{\max}$ on the cross-validated drawdown is tightened, against the equal-weight ensemble (grey lines). Because drawdown is forecastable (Spearman r_S=0.37), tightening $\rho_{\max}$ reduces realised mean and worst-case drawdown monotonically below the equal-weight ensemble.

from the same family can be highly correlated even with different parameters. We measure diversity on training-period returns—available before any selection—with three statistics: the mean pairwise return correlation, split into within-family and across-family pairs; the Jaccard overlap of invested days; and the *effective number of bets* (ENB) $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ computed from the eigenvalues λ_i of the candidates' return-correlation matrix [62], [63], which equals K for uncorrelated and 1 for identical candidates.

Findings. The concern is well founded. In the rolling pools, where two candidates per family are generated, within-family pairs have a mean correlation of 0.80 against 0.33 for across-family pairs (difference +0.46, Wilcoxon $p<10^{-100}$; Fig. 11b), so a second candidate from the same family adds little independent information. Diversity therefore comes mainly from *family coverage*, which our generator enforces by design: the GPT-5.4 prompt requires exactly one candidate per family, and the resulting pools have a mean correlation of 0.37, an invested-day overlap of 0.27, and 2.85 effective bets out of 5 (range 2.63–3.08 across assets; Fig. 11a). The most correlated families are MA crossover and price momentum (0.71), both trend-following, and the least correlated is RSI mean reversion, the only contrarian family (correlation 0.00–0.29 with the others). The K=10 rolling pools yield only 3.39 effective bets, confirming that adding same-family candidates yields diminishing diversity.

Does diversity need to be enforced at selection time? We tested two further safeguards. First, a *family-diverse* top-3 variant that forbids two ensemble members from the same family leaves the rolling Sharpe statistically unchanged (0.75 vs. 0.77, p=0.17); the cross-validated score already spreads the selected members over 2.17 families on average. Second, across cells the ENB of the pool is not associated with the size of the ensemble benefit (Spearman 0.04, p=0.25). Diversity at the generation stage is therefore the effective lever. We ensure it by (i) prompting for one candidate per family, (ii) auditing the ENB of every pool before selection (it is logged with the consequence vectors), and (iii) recommending that additional candidates come from *new* families or contrarian rules rather than from re-parameterising an existing one, since the ENB, not K, determines the effective size of the candidate set in Proposition 1.

N. Selection-Criterion Sensitivity

Fig. 12 reports selection changes under alternative metrics. Calmar-ratio selection agrees with Sharpe-ratio selection on all 10 stocks (**0/10 changes**). Cumulative-return selection differs for 1 stock (**1/10 changes**, AAPL only). Maximum-drawdown minimisation differs for 9 of the 10 stocks (all except NVDA)—expected, as it pursues a qualitatively distinct objective. The 100% agreement between Sharpe and Calmar selection confirms that $\mathrm{CA}^2\mathrm{I}$'s conclusions are robust to the specific choice of risk-adjusted criterion.

O. Limitations

Statistical power and the base vs. enhanced selector. With n=10 assets and a 64-day window, the single-window comparisons lack the power to establish significance, and the *base* in-sample selector is statistically indistinguishable from random even when pooled across the 80-pair rolling analysis (Sections VI-F, VI-H). The enhanced consequence model $\mathrm{CA}^2\mathrm{I}^+$ (Section VI-J) does reach significance on the pooled multi-regime test (Wilcoxon p=0.011; bootstrap CI excludes zero; positive in 10/10 seeds), and its active-capital and risk-budgeted variants significantly beat the equal-weight ensemble on return, Sharpe, and drawdown (Section VI-K). Even so, we are careful not to over-claim: the gain is regime-*aggregate* and does not hold in every individual quarter (notably the 2025Q1 correction). A larger asset universe and multi-year windows would further strengthen the evidence; we report what the present data support.

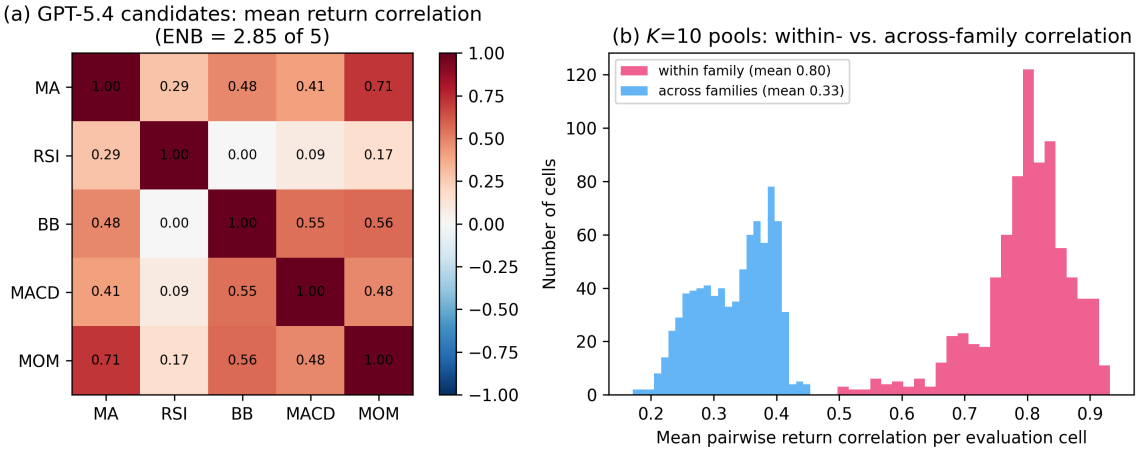


Fig. 11. Candidate diversity on training-period returns. (a) Mean return correlation between the five GPT-5.4 strategy families, averaged over the ten assets (effective number of bets, ENB, 2.85 of 5). (b) Distribution over the 800 rolling cells (80 cells × 10 seeds, K=10) of the mean within-family (pink) and across-family (blue) pairwise correlation: candidates from the same family are strongly correlated (0.80), candidates from different families much less so (0.33).

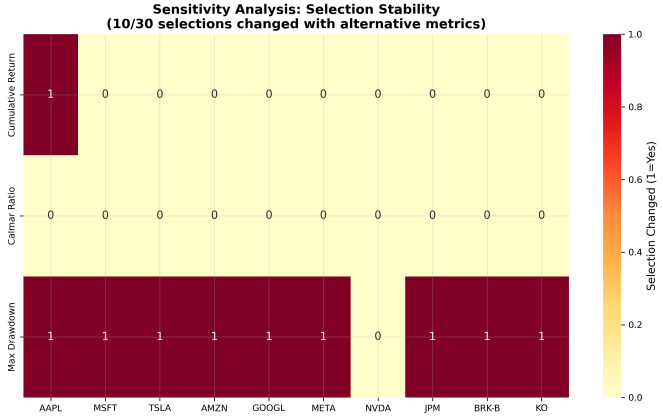


Fig. 12. Sensitivity heatmap: strategy selection stability under alternative evaluation metrics. Dark cells indicate selection changed vs. Sharpe baseline. Sharpe and Calmar agree on 100% of selections; only the qualitatively distinct max-drawdown criterion drives significant changes.

Sensitivity to the consequence model and non-stationarity. CA^{2}I is only as good as its historical consequence model. The stationarity assumption of Theorem 1 is violated in practice; the 2025Q1 window (Section VI-H) and the AAPL/AMZN cases show that a regime shift can make the in-sample-best candidate the out-of-sample worst, so selection can underperform random. Adaptive consequence models with rolling re-estimation, regime detection, and ensembling over windows are the natural remedy.

Transaction costs. Although the headline comparison is frictionless, Section VI-G shows graceful, order-preserving degradation up to 50 bps per side and that CA^{2}I does not select higher-turnover strategies. Costs nonetheless narrow the (already modest) active-vs-passive gap and would matter more for higher-frequency instantiations.

Scope of the instantiation. The study uses long-only, binary-position, single-asset strategies from five technical families. Portfolio-level multi-asset selection, short and leveraged positions, and richer candidate families are important extensions; the framework itself is agnostic to these choices.

Data. The ten assets are large, liquid survivors of the sample period, so results do not speak to small-capitalisation or delisted stocks, and returns are price returns without reinvested dividends (Section V-A); both choices apply identically to every method.

What the consequence model forecasts and how execution matters. Our evidence shows that CAM's strongest single-candidate signal is its cross-validated *drawdown* forecast (r_S=0.37, positive in 73/80 cells), whereas single-candidate quarterly Sharpe rankings are noisy across regime shifts. When paired with active-capital execution (CA^{2}I$^+_{\text{act}}$), consequence-based top-m selection significantly outperforms the static $1/K$ equal-weight ensemble on Sharpe (0.92 vs. 0.69, p=0.015) and annualised return (42.2% vs. 21.7%, p=2.5×10^{-6}) by removing idle cash drag while retaining diversification; when paired with drawdown-aware ranking or a hard risk budget ($\rho_{\max}$=7.5%), it cuts mean drawdown from −5.2% to −3.4% (without cash abstention, p=2.4×10^{-12}) and −2.9% (with $\rho_{\max}$, p=5.8×10^{-9}).

Text-based and neural LLM-as-a-judge rankers. In Section VI-L we evaluate both deterministic text rankers and a local open-weight neural LLM-as-a-judge (`Llama-3.1-8B-Instruct`). Larger frontier judge models could potentially score textual descriptions more nuancely in benign settings, but because any text-only judge conditions on the candidate's self-description rather than simulated execution, it remains structurally exposed to the relabelling attack of Section VI-L while incurring additional token and latency overhead.

VII. Consequence-Aware AI and Responsible AI

A. Transparency and Explainability

In CA^{2}I, every candidate sequence is generated, evaluated, and ranked before any irreversible action occurs. The full

consequence vector $\phi(\tau_k)$ is logged for all K candidates—not only the selected one. For any executed sequence τ^*, an auditor can therefore answer: (i) what alternatives were considered? (ii) what were their predicted consequences? (iii) why was τ^* preferred according to (8)? Vanilla agentic systems cannot answer any of these questions, because they do not generate or evaluate alternatives before acting.

B. Accountability

Because the selection criterion (8) is an explicit mathematical function of the consequence vector, accountability is formally traceable. If an executed strategy causes harm, an auditor can verify whether the harm was foreseeable given the historical consequence model (i.e., $\widehat{\mathrm{DD}}_{k^*}$) or resulted from an unforeseen regime change beyond the model's representational capacity. This distinction—between foreseeable and unforeseeable harms—is central to legal and regulatory accountability under the EU AI Act [7].

C. Harm Avoidance via Hard Risk Constraints

The drawdown constraint $\widehat{\mathrm{DD}}_k \leq \rho_{\max}$ implements formal harm avoidance at the architectural level. Critically, this constraint operates on the *consequences* of proposed sequences—not their textual descriptions. Unlike guardrails applied to LLM output (which can be bypassed by adversarial prompts that describe a harmful strategy as benign), CAM evaluates strategies by *simulating* them. A strategy that describes itself as low-risk but exhibits large drawdown under simulation is filtered regardless of its description. Section VI-L verifies this experimentally: relabelling the riskiest proposal as conservative captures a text ranker in every cell but leaves CAM's decisions unchanged. The risk-budget study of Section VI-K shows, moreover, that the constraint is not merely formal: because simulated drawdown forecasts realised drawdown, tightening $\rho_{\max}$ reduces realised mean and worst-case drawdown monotonically.

D. Human Oversight and Intervention

CA^{2}I's architecture naturally supports human-in-the-loop oversight. The consequence vectors $\{\phi(\tau_k)\}_{k=1}^{K}$ can be surfaced to a human supervisor before execution. The supervisor can override $\mathcal{S}$, choose a different τ_k from the evaluated set, or reject all candidates and trigger re-generation. This contrasts with vanilla agents, where the human must evaluate raw action sequences without any quantitative consequence estimate.

E. Generalisation to Other High-Stakes Domains

While instantiated in finance, CA^{2}I is domain-agnostic. In *clinical decision support* [3], CAM could simulate treatment pathways against electronic health records to estimate adverse-event rates before a protocol is recommended. In *autonomous code execution*, CAM could run candidate code in a sandbox to detect sequences that consume excessive resources or modify protected files. In *supply chain management*, CAM could simulate logistics sequences against historical demand data to estimate stockout and overstock risks. In each case, the architectural pattern is identical: generate $\mathcal{T}$, assess $\{\phi(\tau_k)\}$, select τ^*, execute.

VIII. Conclusion

We presented Consequence-Aware Agentic AI (CA^{2}I), a formal framework that augments LLM agents with a pre-execution Consequence Assessment Module. CA^{2}I decouples generation from execution, inserting a principled simulation-and-selection step that evaluates the predicted consequences of all candidate sequences before any irreversible action is taken.

Theoretically, we proved that risk-adjusted selection dominates random selection in expectation under stationarity (Theorem 1); quantified the dominance gap and showed it grows with the candidate-set size K (Proposition 1); and formalised maximum drawdown as a hard architectural risk constraint (equation (5)).

Empirically, on ten diversified US equities we found—reported candidly—that CA^{2}I attains a higher mean out-of-sample Sharpe (0.51) than uniform random selection (0.37) and than three established portfolio-optimisation selectors (mean-variance 0.19, risk-parity 0.45, minimum-variance -0.63), and reduces drawdown by 18% relative to buy-and-hold, while *not* increasing turnover and degrading gracefully under transaction costs. We also found, and disclose, that single-window pairwise differences among the active selectors are not statistically significant at $n{=}10$ (bootstrap CIs include zero), and that the *base* in-sample selector is statistically indistinguishable from random when pooled across a multi-regime rolling evaluation. This diagnosis motivated our enhanced consequence model, $\mathbf{CA^2I^+}$ (walk-forward cross-validation, a robustness-penalised score, and a top-m ensemble), which *does* achieve a statistically significant, regime-robust edge: pooled over eight quarterly windows ($n{=}80$) it lifts mean out-of-sample Sharpe to 0.77 versus 0.53 for random ($+0.24$; Wilcoxon $p{=}0.011$; bootstrap CI $[+0.015, +0.45]$ excluding zero; positive in all 10 generation seeds) and beats the base selector by $+0.21$, while reducing drawdown relative to both. Performance also rises with K as the theory predicts, and Sharpe and Calmar selection agree on 100% of choices.

A comprehensive comparison with the equal-weight (EW) ensemble and generate-and-rank pipelines sharpens what these gains mean. First, because individual technical strategies hold positions on only 41.5% of trading days, static $1/K$ equal weighting leaves nearly 60% of capital idle in cash; pairing consequence-ranked top-m selection with active-capital execution ($\mathrm{CA^2I^+_{act}}$) significantly beats EW on rolling Sharpe (0.92 vs. 0.69, Wilcoxon $p{=}0.015$), Sortino (1.78 vs. 1.50, $p{=}0.020$), and annualised return (42.2% vs. 21.7%, $p{=}2.5{\times}10^{-6}$), while lifting deployment Sharpe from 0.57 (EW) to 0.95 and deployment annualised return from 32.2% to 77.0%. Second, CAM's cross-validated drawdown estimate strongly forecasts realised out-of-sample drawdowns (mean Spearman $r_S{=}0.37$, positive in $73/80$ cells): ranking by cross-validated drawdown without cash abstention cuts rolling mean drawdown from -5.2% (EW) to -3.4% ($p{=}2.4{\times}10^{-12}$), and enforcing a hard per-proposal budget ($\rho_{\max}{=}7.5\%$) cuts mean drawdown to -2.9% ($p{=}5.8{\times}10^{-9}$) and eliminates all cells exceeding 10% drawdown at statistically unchanged Sharpe. Third, against both deterministic text rankers and a neural LLM-as-a-judge (`Llama-3.1-8B-Instruct`), simulation-

based assessment delivers significantly shallower drawdowns (-5.8% vs. -7.6% for the neural judge, $p{=}3.4{\times}10^{-5}$), is completely immune (0% change) to an adversarial relabelling attack that steers both the lexicon ranker and the neural LLM judge to the riskiest candidate in 100% of cells, and costs under 0.04 s of CPU time and zero LLM tokens per candidate (versus $\approx$ 1,000–1,300 tokens per cell for the neural judge). Diversity among candidates comes from family coverage, since same-family candidates are highly correlated (0.80), and is best ensured at generation time.

From a Responsible AI perspective—which we regard as the framework's principal contribution—CA^{2}I provides concrete, computationally tractable mechanisms for transparency, accountability, and harm avoidance, as required by the EU AI Act, the NIST AI RMF, and IEEE Ethically Aligned Design. Because the risk filter operates on *simulated consequences* rather than textual descriptions, it is more resistant to prompt manipulation than purely textual guardrails, as our relabelling experiment confirms.

Future work will extend CA^{2}I to multi-period adaptive consequence models, portfolio-level multi-asset selection, non-financial agentic domains including clinical decision support and autonomous code execution, and causal consequence models that reason about counterfactual outcomes beyond historical simulation.

ACKNOWLEDGMENT

The authors acknowledge the open-source `yfinance` community for maintaining free access to historical financial data. The style of this article has been polished using *QuillBot*.

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